JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B3, PAGES 3979–4006, MARCH 10, 1991

Paleomagnetic Evidence for the Age and Extent of Middle Tertiary Counterclockwise Rotation, Dixie Valley Region, West Central Nevada

MARK R. HUDSON

U.S. Geological Survey, Denver, Colorado

JOHN W. GEISSMAN

Department of Geology, University of New Mexico, Albuquerque

Paleomagnetic data obtained from Oligocene to lower Miocene igneous rocks and middle Miocene basaltic rocks at fifteen localities from a region surrounding Dixie Valley in west central Nevada indicate that parts of the area experienced counterclockwise vertical-axis rotation, and these data provide constraints on the extent and timing of rotation. Counterclockwise vertical-axis rotation probably exceeding 30° is indicated for Oligocene to lower Miocene rocks in the central part of the study area. Calculated rotations increase with increasing age of the Oligocene to lower Miocene rocks, indicating that vertical-axis rotation was concurrent with ash-flow deposition (about 23–33 Ma). Paleomagnetic data indicate that middle Miocene basaltic rocks (about 10–17 Ma) postdate significant counterclockwise vertical-axis rotation, although this interpretation is complicated because the rocks were erupted episodically in pulses that apparently were short lived with respect to geomagnetic secular variation. Counterclockwise rotation was related to a deformation event that predated development of the present basin and range physiography of the area. The total amount of crustal rotation in Tertiary time in the area is poorly known. Rotation estimates calculated from time-averaged mean directions that incorporate data from a broad range of the Oligocene to lower Miocene units probably underestimate total Tertiary rotation because the mean directions include data from units that postdate much of the rotation. For example, a rotation estimate for a composite sequence from the central part of the study area is $-23° \pm 15°$, whereas estimates from older and younger halves of the sequence are $-37° \pm 21°$ and $-11° \pm 16°$, respectively. Paleomagnetic data indicate that Oligocene to lower Miocene rocks at some localities in the northern and southern parts of the study area (e.g., the Golconda Canyon locality) probably did not experience significant Tertiary counterclockwise rotation. Although the paleomagnetic data alone are insufficient to fully define the geographic area subjected to Tertiary counterclockwise rotation, the data suggest that a discrete subregion was affected rather than the entire Basin and Range province.

INTRODUCTION

Rigid-body rotation of rocks is an important component of deformation in the brittle upper crust. Rigid-body rotation and fault translation can work in concert to efficiently accommodate large strains [*Thompson*, 1960]. Rotations about horizontal and vertical axes are common end members of geometrically possible rigid-body rotations. Rotations about inclined axes also occur, but they are difficult to document, and in practice, such rotations often are factored into components of horizontal- and vertical-axis rotation. Dipping stratification observed in upper crustal rocks indicates that rotation about a horizontal axis (tilting) is virtually ubiquitous in orogenic belts. Detection of rotation about a vertical axis is more difficult because few rocks contain unambiguous markers that allow detection and quantification of vertical-axis rotation. In appropriate rocks, directions of remanent magnetization can be used to detect vertical-axis rotation. Paleomagnetic studies indicate that vertical-axis rotation is common in orogenic belts affected by strike-slip faults [e.g., *Beck*, 1980; *Luyendyk et al.*, 1980; *Walcott*, 1984; *Lamb*, 1988]. Vertical-axis rotation in extensional terranes is also well documented [e.g., *Brown and Golombek*, 1985; *Hudson and Geissman*, 1987; *Kissel and Laj*, 1988; *Ross et al.*, 1989; *Wells and Hillhouse*, 1989; *Rosenbaum et al.*, 1991]. Recognition and quantification of vertical-axis rotation provides important information on the kinematics of extensional terranes.

This paper presents paleomagnetic data from Tertiary igneous rocks in a region surrounding Dixie Valley in west central Nevada (Figure 1). Earlier investigations of Oligocene to lower Miocene rocks in part of the area detected counterclockwise vertical-axis rotation [*Hudson and Geissman*, 1987]. In this paper we constrain the vertical-axis rotation to predate middle Miocene time based on paleomagnetic data from middle Miocene basaltic rocks in the area and by a more detailed investigation of Oligocene to lower Miocene rocks at additional localities. We also partly constrain the geographic extent of the structural domain affected by vertical-axis rotation from results of expanded sampling of Oligocene to lower Miocene rocks in the area. Recognition of vertical-axis rotation has implications for the style of deformation in the area during Oligocene to early Miocene time, and it is also important in assessing the original geometry of structures associated with older Mesozoic tectonic events [e.g., *Speed*, 1983; *Oldow*, 1984] in the region.

GEOLOGIC SETTING

Tertiary volcanic and nonmarine sedimentary rocks in the study area (Figure 1) overlie various Upper Paleozoic and Mesozoic metasedimentary and igneous rocks, often in

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Paper number 90JB02424.
0148-0227/91/90JB-02424$05.00

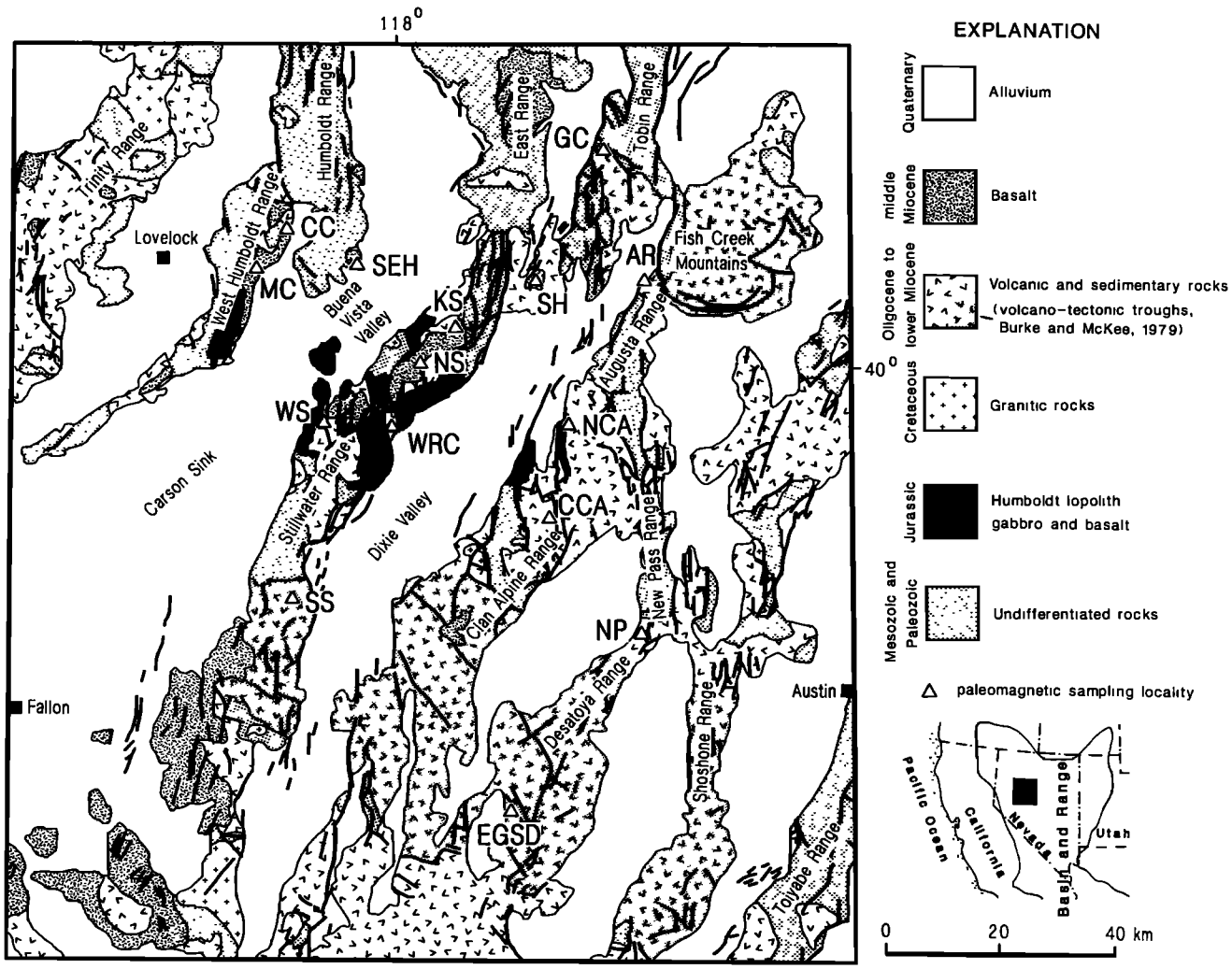


Fig. 1. Generalized geologic map of the region surrounding Dixie Valley in west central Nevada (modified from *Stewart and Carlson* [1978] and *Burke and McKee* [1979]) showing localities of samples collected for paleomagnetic studies. AR, Augusta Range; CC, Coal Canyon; CCA, central Clan Alpine Range; EGSD, Eastgate-southern Desatoya Range; GC, Golconda Canyon; KS, Kitten Springs; MC, Muttlebury Canyon; NCA, northern Clan Alpine Range; NP, New Pass; NS, northern Stillwater Range; SEH, southeastern Humboldt Range; SH, Sou Hills; SS, southern Stillwater Range; WRC, White Rock Canyon; WS, western Stillwater Range.

angular unconformity [*Speed*, 1976; Willden and Speed, 1974]. Locally preserved lower Oligocene andesite lavas are overlain by an extensive series of Oligocene to lower Miocene ash-flow tuffs. K-Ar ages of the andesites are 36–33 Ma [*Riehle et al.*, 1972; *Burke and McKee*, 1979], and K-Ar ages of the tuffs range from 33 to 23 Ma [*McKee and Stewart*, 1971; *Riehle et al.*, 1972; *Silberman and McKee*, 1972; *Speed*, 1976; *Burke and McKee*, 1979] (when applicable, cited K-Ar ages are adjusted for new constants [*Dalrymple*, 1979]). Two elongate volcanic depressions (Figure 1), termed volcano-tectonic troughs by *Burke and McKee* [1979], lie in northern and southern parts of the area. Ash-flow tuffs in the troughs are thick and densely welded, and many may have accumulated in intracaldera settings. Intrusive rocks are also common within the southern trough. Ash-flow tuffs in the central area between the troughs are thin, variably welded, and widely distributed, and they were probably deposited as outflow facies of ash-flow eruptions. Few geologic studies have subdivided the ash-flow tuff sequences in the area, and individual tuffs have not been correlated among ranges. Some tentative correlations of units between ranges are made in this study on the basis of lithology and magnetostratigraphy. In some ranges, Miocene nonmarine sedimentary rocks overlie the Oligocene to lower Miocene volcanic rocks [*Page*, 1965; *Hastings*, 1979; *VanLandingham*, 1988].

Middle Miocene flows of basalt and basaltic andesite (hereinafter referred to as basalt) and subordinate pyroclastic deposits are present in many ranges. K-Ar ages from the basalt sequence range from 8 to 17 Ma [*Hastings*, 1979; *Nosker*, 1981]. In separate areas the middle Miocene basalts directly overlie Mesozoic rocks, Oligocene to lower Miocene volcanic rocks, or Miocene nonmarine sedimentary rocks. Upper Miocene(?) to Quaternary alluvial, fluvial, and lacustrine strata overlie the basalt sequence in the present-day basins [*Hastings*, 1979; *Okaya and Thompson*, 1985].

We recognize two temporally discrete phases of deformation that affected the area in Tertiary time. The older deformation affected, and locally it can be shown was coeval with, Oligocene to lower Miocene rocks in parts of the area, but it apparently predated deposition of the middle Miocene basalt sequence. Knowledge about the older deformation phase, however, is sparse due to a lack of detailed studies. In the

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Tobin Range, *Burke* [1977] mapped generally west-northwest dipping normal faults that in some cases offset but in other cases are overlapped by moderately east dipping Oligocene to lower Miocene volcanic rocks. In the central Stillwater Range, Oligocene to lower Miocene rocks are in places moderately to steeply tilted to the northeast or southwest and cut by north to northwest striking faults that are overlapped by nearly flat-lying middle Miocene basalt flows [*Hudson and Geissman*, 1987; *Hudson*, 1988]. Geologic relations indicate that some of these northwest trending faults had right-lateral offset and that they were active during deposition of the Oligocene to lower Miocene sequence. In the southern Stillwater Range, Oligocene to lower Miocene ash-flow tuffs dip steeply both to the west and to the east, but the west flank of the range is overlapped by gently west dipping middle Miocene basalts [*Page*, 1965; D. A. John, unpublished data, 1989]. Data presented in this paper illustrate that vertical-axis rotation affected part of the area during this older phase of Tertiary deformation.

The second phase of deformation in the area was a late Miocene to Quaternary episode of normal faulting that produced the present physiography of the area (see reviews by *Okaya and Thompson* [1985] and *Anderson et al.* [1983]). Geophysical and geologic studies have shown that in Dixie Valley this youngest deformation postdates the middle Miocene basalt sequence, and it was characterized by high-angle normal faulting and associated horst and graben formation with only mild tectonic tilting.

Methods for Paleomagnetic Data Collection

Samples for paleomagnetic analysis were collected at fifteen localities from the Stillwater, Clan Alpine, Tobin, Augusta, Humboldt, West Humboldt, New Pass, and Desatoya Ranges and the Sou Hills (Figure 1). Samples (1679) from 226 sites from Tertiary rocks in the area were analyzed. At individual sites typically six to ten samples were collected from a single geologic unit over an area of several tens of square meters. All samples were collected with a portable drill and oriented using both magnetic and sun compasses. Attitudes of the strata used for tilt-correction of the paleomagnetic data were estimated from compaction foliations or boundaries of welding zones in tuffs, boundaries of flows in lavas, or bedding of sedimentary strata intercalated with the volcanic units.

In the laboratory, remanent magnetizations were measured with a Schonstedt DSM-1 spinner magnetometer. Alternating field (AF) and thermal demagnetization of specimens were performed using Schonstedt GSD-5 and TSD-1 instruments, respectively. For most sites, detailed pilot demagnetization (10–15 steps) on one or two specimens was followed by demagnetization of the remaining samples, using 6–9 steps. AF demagnetization principally was used because it was effective in isolating a well-defined and well-grouped magnetization in most volcanic rocks. Representative samples from many sites were thermally demagnetized for comparison with AF results and to assess unblocking temperature spectra of the units. For the IXL pluton from the southern Stillwater Range and a few other sites in volcanic rocks, thermal demagnetization was the most effective technique for isolating components of magnetization. In individual specimens, directions of magnetization components were determined by a least squares fit of linear segments of vector diagrams. Most site means and dispersion parameters were calculated using Fisher statistics [*Fisher*, 1953]. For a few sites, intersections of remagnetization circles [*Halls*, 1976] were used to derive site mean directions. Bingham statistics [e.g., *Onstott*, 1980] were used to assess confidence limits for means derived from remagnetization circles. Angular standard deviation of virtual geomagnetic poles was calculated as by *Doell* [1970] to provide a measure of secular variation.

Rock magnetic properties of representative samples were investigated using thermomagnetic analysis and acquisition and direct-field demagnetization of isothermal remanent magnetization (IRM). Petrographic inspection of polished thin sections also supplemented information gained from demagnetization behavior.

Tests for Rotation

Paleomagnetic data provide reference directions of two types that potentially can be used to detect and quantify tectonic rotation. A single widespread volcanic unit may cool quickly and record an instantaneous geomagnetic field direction that can be used to accurately detect relative rotation among locations [e.g., *Magill et al.*, 1982; *Hagstrum and Gans*, 1989; *Wells and Hillhouse*, 1989; *Rosenbaum et al.*, 1991]. Although this method is typically the most sensitive, it is not generally applicable in the Dixie Valley region because individual cooling units have not been correlated unambiguously to areas that are known not to have been rotated with respect to the craton. Nevertheless, comparison of unit directions of individual tuffs provides a measure of differential rotation among localities within the region. More commonly, study of all available cooling units at a locality allows an estimate of the time-averaged geomagnetic field direction. When compared to a time-averaged axial dipole field direction, discordance between observed and expected directions in some cases can be attributed to rotation of the locality with respect to the craton.

Paleomagnetic data used to assess vertical-axis rotation in this study are from two time sequences of rocks: (1) Oligocene to early Miocene and (2) middle Miocene. For each sequence, data from individual localities are discussed and, when four or more unit means were available, a time-averaged locality mean direction is calculated for comparison with an expected direction. The angular difference between declinations of tilt-corrected observed and expected directions (R value, *Beck* [1980], with statistical limits from *Demarest* [1983]) is an estimate of vertical-axis rotation. The angular difference between inclinations of expected and tilt-corrected observed directions (F value, *Beck* [1980], with statistical limits from *Demarest* [1983]) is also calculated for inspection. Expected directions for the area are calculated from the 30-Ma and 10-Ma pole positions of *Irving and Irving* [1982] for comparison with data from the Oligocene to lower Miocene and middle Miocene rocks, respectively. An expected direction calculated from the 38- to 22-Ma pole position of *Diehl et al.* [1983] lies within 2° of that calculated from the 30-Ma pole of *Irving and Irving* [1982], and the two poles are not statistically distinct at a 95% confidence level. Therefore conclusions of this paper are not strongly affected by the choice of reference pole.

Oligocene to Lower Miocene Tuffs and Related Rocks

General Characteristics

Rocks sampled from the Oligocene to lower Miocene sequence are predominantly ash-flow tuffs, but andesitic

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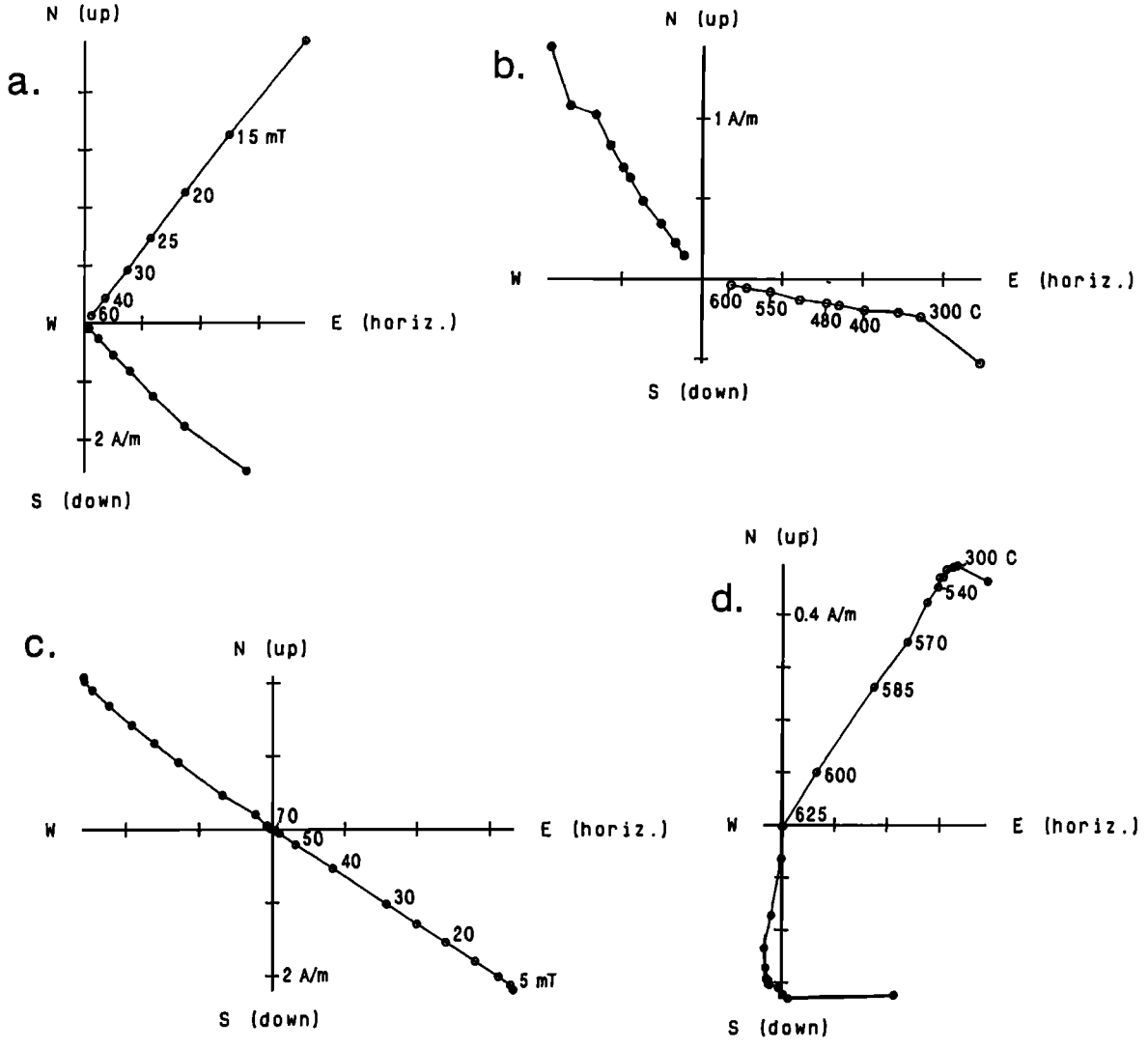


Fig. 2. Representative orthogonal demagnetization diagrams, shown in in situ coordinates, from Oligocene to lower Miocene volcanic rocks from the Dixie Valley region. Solid and open circles are projections on horizontal and vertical planes, respectively. Demagnetization levels are given in milliTesla (mT) or degrees Celsius (C). (*a*) Devitrified welded ash-flow tuff, site 306, western Stillwater locality. (*b*) Basalt flow, site 325, Golconda Canyon locality, Tobin Range. (*c*) Devitrified welded ash-flow tuff, site 160, northern Clan Alpine Range locality. (*d*) Devitrified welded ash-flow tuff, site 548, central Clan Alpine Range locality.

lavas and a granodiorite pluton were also sampled. Paleomagnetic data and the geologic setting of each locality are discussed briefly. Data from *Hudson and Geissman* [1987] are included to more completely evaluate the suitability of these data for detection of vertical-axis rotation.

After removal of a small-moment component that was probably a viscous remanent magnetization (VRM) at low demagnetization levels, most samples from the Oligocene to lower Miocene rocks contained a single component of remanent magnetization that was well defined in both AF and thermal demagnetization (Figure 2). Within a site, these components typically are well grouped (Figure 3); their site means typically have k values [*Fisher*, 1953] greater than 100 (Table 1). Samples from a few sites contain more complex magnetization. Data from these sites are discussed in descriptions of the individual localities. Where multiple sites were collected from an individual cooling unit at a locality, their tilt-corrected means were averaged to derive a unit mean (Table 1, Figure 3).

Localities

White Rock Canyon area, Stillwater Range. The White Rock Canyon area lies on the eastern flank of the Stillwater Range (WRC, Figure 1). Geologic relations among the units at this locality are relatively well known [*Hudson*, 1988]. At this locality, an approximately 1-km-thick sequence of ash-flow tuffs is exposed in a series of fault blocks (Figure 4). Angular unconformities between some units and the presence of faults that offset older units but that are overlapped by younger units indicate that deformation was concurrent with ash-flow deposition. For paleomagnetic analysis, 13 separate cooling units were sampled from the ash-flow tuff sequence. The youngest ash-flow tuff of the sequence is a distinctive crystal-rich rhyolite correlated with the 23-Ma New Pass Tuff [*McKee and Stewart*, 1971] on the basis of similar lithology, relative stratigraphic position, and direction of remanent magnetization [*Hudson and Geissman*, 1988]. Biotite from biotite-rich tuff in the middle part of the overall sequence yielded a K-Ar age of

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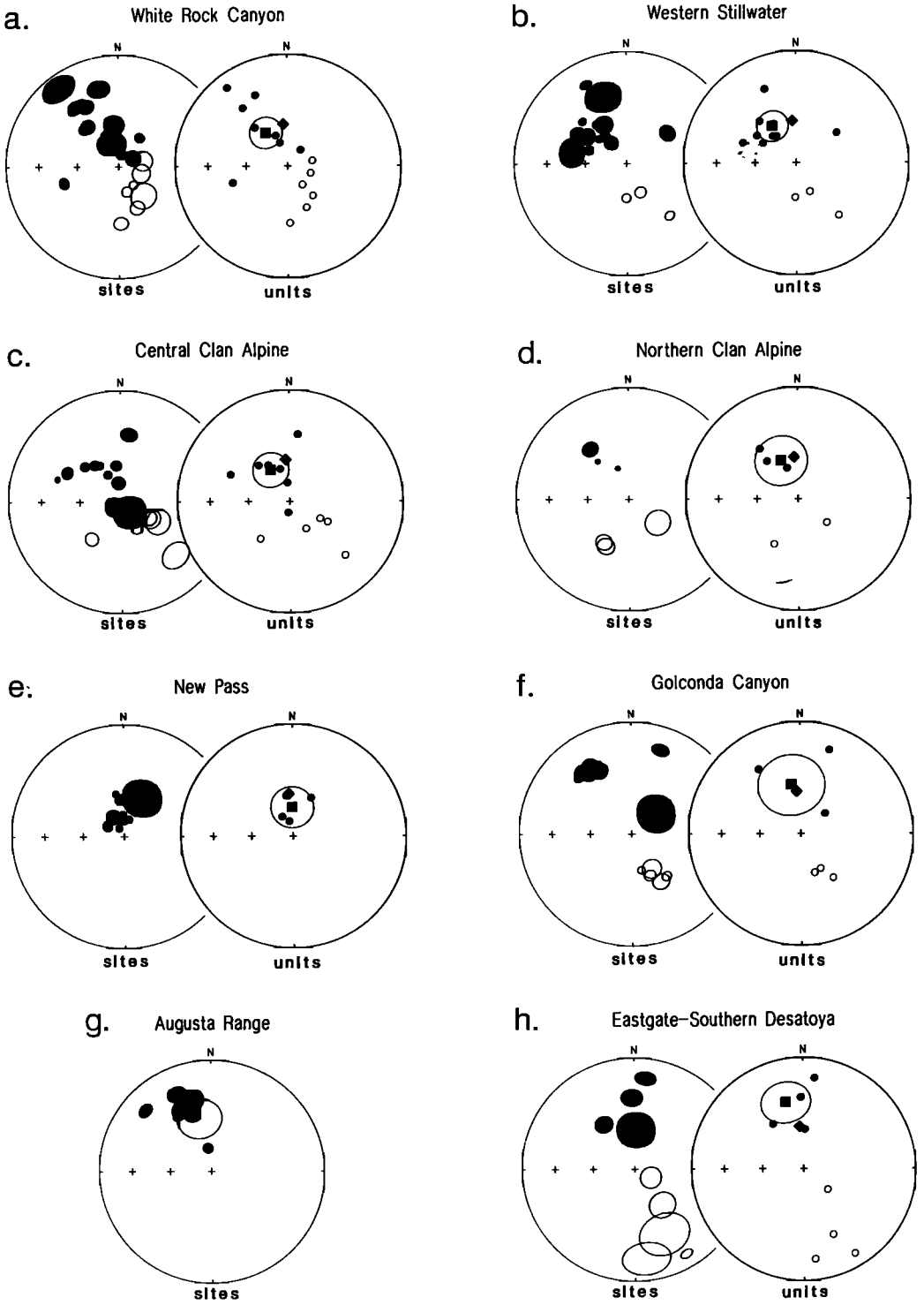


Fig. 3. Equal-area projections summarizing paleomagnetic data obtained at localities from Oligocene to lower Miocene volcanic rocks. Solid and open symbols are projections on lower and upper hemispheres, respectively. For each locality the left projection shows the cones of 95% confidence about all tilt-corrected site means, and the right projection shows tilt-corrected unit means (circles), the locality mean (square with cone of 95% confidence), and the expected time-averaged direction (diamond). Numerous andesite flows at the Augusta Range locality were probably remagnetized. Thus, cones of 95% confidence about site means for this locality are shown in in situ coordinates (Figure 3*g*), and no locality mean is calculated.

27.2 + 0.7 Ma (D. A. John and E. H. McKee, unpublished information, 1989). The tuff sequence was intruded by several diabase dikes and sills and capped by a thin (about 20 m) basalt unit. Two sills and the capping basalt were also sampled for paleomagnetic analysis. Geologic relations indicate that the diabase dikes and sills are equivalent in age to the upper parts of the ash-flow sequence [*Hudson*, 1988]. A total of 30 sites from 15 separate units were analyzed from this locality.

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TABLE 1. Paleomagnetic Data From Oligocene to Lower Miocene Rocks From the Region Surrounding Dixie Valley, West Central Nevada

Number	Site		N/N_0	D_0	I_0	a_{95}	k	D_1	I_1	D_2	I_2	VGP	
	Latitude	Longitude										Latitude	Longitude
						White Rock Canyon (WRC)							
378	39.905	−118.028	8/8	70	−50	7.0	63	77	−72	77*	−72	−27	26
379	39.906	−118.028	7/7	84	−54	6.6	84	107	−73	107*	−73	−42	20
380	39.907	−118.028	7/7	111	−51	9.3	43	140	−62	140*	−62	−60	352
381	39.907	−118.028	7/7	339	33	7.0	77	346	29	336*	31	61	114
242	39.897	−118.018	8/8	330	35	4.7	142	329	36				
495	39.908	−118.007	8/10	359	31	5.2	117	332	39				
383	39.911	−118.029	6/6	341	39	7.0	93	354	59	354*	59	85	149
243	39.898	−118.019	9/9	144	−74	3.0	300	123	−74	142*	−73	−60	22
382	39.911	−118.028	7/7	150	−49	3.7	262	162	−71				
596	39.902	−118.008	10/10	287	76	3.7	175	253	48	253*	48	7	184
225	39.909	−118.012	8/8	12	50	3.7	229	339	66	339*	66	73	187
224	39.908	−118.013	8/8	328	11	10.2	31	323	13	323*	13	43	117
387	39.925	−118.027	6/7	186	−47	4.9	190	178	−48	178*	−48	−79	252
384	39.922	−118.029	7/7	165	−54	5.1	141	156	−57	156*	−57	−71	332
385	39.921	−118.025	6/7	325	52	7.4	82	334	72	342*	72	69	214
388	39.925	−118.028	7/7	13	72	7.9	60	351	73				
490	39.913	−118.009	9/9	343	44	5.1	102	346	50†				
386	39.921	−118.024	7/7	351	70	5.5	120	60	78	38*	75	59	276
389	39.926	−118.029	7/7	43	77	3.3	327	14	81				
488	39.913	−118.008	8/8	23	62	3.4	260	36	64				
489	39.913	−118.008	5/8	84	57	15.7	25	92	53†				
218	39.902	−118.011	4/4	324	62	6.5	201	292	55†				
219	39.903	−118.012	4/4	342	55	6.7	191	312	56†				
220	39.903	−118.012	4/4	331	63	8.6	116	295	58†				
223	39.908	−118.014	4/4	340	68	4.2	483	293	65†				
491	39.912	−118.010	8/8	346	39	4.0	190	323	35	323*	35	52	131
493	39.912	−118.010	8/8	348	39	4.8	136	324	35				
494	39.913	−118.010	6/6	345	41	3.2	437	321	36				
495	39.922	−118.025	8/8	334	53	3.8	217	318	53	320*	53	58	154
496	39.922	−118.025	7/7	339	52	5.0	145	323	53				
						Western Stillwater Range (WS)							
300	39.901	−118.166	8/8	324	6	3.4	262	333	22	337	29	58	108
301	39.900	−118.166	8/8	330	27	11.8	23	341	37				
302	39.900	−118.165	6/6	124	−50	3.2	449	181	−65	182*	−65	−83	72
317	39.901	−118.163	5/6	129	−34	3.3	539	141	−40	141*	−40	−53	317
303	39.900	−118.165	8/8	288	24	5.6	97	305	55	305	55	47	164
304	39.899	−118.165	8/8	289	17	4.9	128	301	48	291	48	34	164
305	39.899	−118.165	8/8	275	14	9.5	35	281	49				
306	39.898	−118.166	8/8	129	−53	4.0	195	155	−67	155	−67	−70	11
307	39.898	−118.165	8/8	303	50	3.3	283	322	66	324*	66	63	182
308	39.897	−118.165	8/8	308	44	2.7	430	325	60				
309	39.896	−118.165	8/8	299	56	4.5	150	324	72				
310	39.897	−118.164	8/8	25	65	5.6	98	53	53	53*	53	48	323
311	39.893	−118.158	7/8	293	30	2.6	560	314	45	319	49	55	149
314	39.898	−118.160	8/8	304	47	2.9	369	323	54				
312	39.893	−118.157	9/9	277	44	2.4	475	312	63	301	62	46	177
313	39.894	−118.157	8/8	268	37	3.2	295	290	61				
315	39.900	−118.161	7/7	306	52	6.3	93	330	57	330*	57	67	157
316	39.900	−118.162	9/9	294	55	2.2	544	319	64	319*	64	60	176
						Central Clan Alpine Range (CCA)							
204	39.797	−117.654	7/7	112	−48	9.0	46	119	−59	119	−59	−45	353
205	39.797	−117.653	4/5	131	−22	9.7	91	135	−32	135	−32	−45	315
206	39.797	−117.651	7/7	356	40	5.3	133	7	39	7	39	72	42
207	39.795	−117.648	6/7	321	52	2.9	540	330	59	330	59	67	161
208	39.795	−117.648	7/8	116	−57	7.0	75	120	−65	120	−65	−47	3
209	39.795	−117.647	6/8	115	−57	5.1	171	120	−66				
211	39.795	−117.645	8/8	328	71	4.7	139	353	76	353*	76	67	235
610	39.795	−117.646	8/8	290	31	1.9	874	290	40	295*	42	34	158
210	39.795	−117.646	8/8	297	36	4.1	187	299	45				
212	39.795	−117.644	8/8	138	−60	3.7	221	150	−67	150*	−67	−67	7
213	39.795	−117.642	8/8	310	45	3.2	297	315	53	320	55	58	158
547	39.795	−117.640	8/8	317	50	3.5	245	325	57				
548	39.795	−117.640	9/9	205	−58	4.8	114	218	−55	218*	−55	−60	148
549	39.794	−117.640	7/8	339	59	3.7	267	353	63	344	65	76	191
550	39.794	−117.639	8/8	321	62	2.9	373	335	68				
551	39.794	−117.639	9/9	263	77	6.7	60	234	84	188	82	24	240
552	39.794	−117.638	5/8	205	84	11.8	43	141	80				

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TABLE 1. (continued)

Number	Site		N/N_0	D_0	I_0	a_{95}	k	D_1	I_1	D_2	I_2	VGP	
	Latitude	Longitude										Latitude	Longitude
						Northern Clan Alpine Range (NCA)							
161	39.925	−117.655	5/7	117	−37	9.2	71	130	−63	130	−63	−53	356
160	39.924	−117.655	7/7	307	30	1.7	1310	321	54	321	54	59	155
155	39.924	−117.654	7/7	313	20	5.4	128	323	43	323	43	56	137
156	39.924	−117.654	7/7	171	−53	5.8	109	210	−53	208	−52	−66	162
157	39.924	−117.653	7/7	170	−50	6.1	99	206	−51				
159	39.925	−117.653	12/13	311	46	1.6	759	341	66	341	66	74	191
						New Pass (NP)							
363	39.579	−117.528	7/7	313	57	5.2	115	331	74	331	74	62	211
364	39.571	−117.514	7/7	339	55	2.1	795	349	58	351	60	83	166
365	39.571	−117.514	7/7	343	60	3.5	303	354	63				
371	39.572	−117.513	16/16	13	60	13.7	8	25	59	25	59	71	324
362	39.579	−117.528	8/8	79	87	3.4	262	301	76	346	79	60	232
366	39.572	−117.513	7/7	347	75	2.4	650	15	77				
367	39.572	−117.513	7/7	337	71	2.3	691	357	74				
368	39.572	−117.513	6/6	305	77	2.5	739	328	83				
369	39.573	−117.513	8/8	331	75	3.4	259	357	78				
						Golconda Canyon, Tobin Range (GC)							
325	40.327	−117.603	8/8	327	10	3.3	287	330	29	327	32	53	124
326	40.327	−117.602	8/8	326	9	4.1	183	329	28				
327	40.327	−117.602	8/8	328	19	5.8	91	333	37				
328	40.327	−117.602	8/8	325	12	6.3	79	328	31				
329	40.327	−117.602	8/8	321	18	3.3	286	324	37				
330	40.327	−117.601	8/8	327	14	2.9	361	334	32				
331	40.328	−117.601	8/8	324	14	3.4	264	327	33				
332	40.328	−117.601	8/9	324	17	3.2	295	327	36				
333	40.328	−117.601	7/7	319	13	3.5	291	322	32				
334	40.328	−117.600	7/7	321	12	2.0	885	323	31				
335	40.328	−117.600	7/7	316	12	3.1	392	317	32				
336	40.329	−117.600	8/8	320	12	5.8	91	322	31				
318	40.320	−117.575	8/8	132	−54	2.6	441	166	−63	161	−60	−74	341
319	40.320	−117.575	8/8	132	−46	4.0	196	157	−57				
320	40.320	−117.574	6/7	125	−36	3.3	402	141	−50	144	−50	−59	325
321	40.320	−117.574	5/7	127	−29	5.9	172	149	−49				
322	40.320	−117.570	9/9	306	72	13.6	15	50	67	50	67	54	301
324	40.321	−117.567	6/6	118	−40	6.9	96	151	−61	515	−61	−68	345
323	40.320	−117.566	4/4	2	33	5.2	317	19	22	19	22	57	27
						Augusta Range (AR)							
337	40.125	−117.507	7/8	338	34	4.1	214	10	35				
338	40.125	−117.506	8/8	338	29	2.9	357	5	32				
339	40.125	−117.506	8/8	337	24	4.5	155	359	29				
340	40.125	−117.505	7/7	336	25	5.7	115	0	30				
341	40.125	−117.505	4/4	338	36	4.8	365	11	36				
342	40.125	−117.505	8/8	342	46	5.1	119	24	40				
343	40.125	−117.504	8/8	342	43	5.5	103	20	38				
344	40.126	−117.504	8/8	334	39	6.2	81	12	41				
345	40.126	−117.503	8/8	329	42	3.3	282	356	43				
346	40.126	−117.502	7/7	352	73	3.5	295	44	55				
347	40.126	−117.502	6/7	344	34	8.3	66	3	30				
348	40.126	−117.501	5/6	348	−50	15.9	26	317	−44				
349	40.126	−117.501	8/8	313	21	4.6	149	315	23				
350	40.126	−117.500	7/7	336	28	2.5	590	339	28				
351	40.126	−117.500	6/6	337	30	4.1	265	340	29				
352	40.126	−117.499	7/8	339	36	2.3	662	344	35				
						Eastgate–Southern Desatoya Range (EGSD)							
353	39.309	−117.875	5/8	290	−41	7.6	103	118	−77	131	−67	−54	5
354	39.307	−117.871	7/9	162	−63	9.4	42	143	−57				
355	39.301	−117.862	8/8	353	−25	6.7	70	358	36	358	36	70	68
356	39.296	−117.860	7/7	6	13	6.0	101	7	19	7	19	60	49
357	39.238	−117.786	5/8	174	−43	16.7	22	156	−36	156	−36	−62	296
358	39.243	−117.783	7/7	355	59	6.1	98	326	50	326	50	61	145
359	39.254	−117.749	7/7	168	−31	3.7	268	149	−13	149	−13	−47	290
360	39.264	−117.743	6/10	195	−15	14.4	23	173	−20	173	−20	−60	255
361	39.264	−117.737	5/10	328	54	13.1	35	2	61	2	61	87	271

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TABLE 1. (continued)

Number	Site		N/N_0	D_0	I_0	a_{95}	k	D_1	I_1	D_2	I_2	VGP	
	Latitude	Longitude										Latitude	Longitude
						IXL Pluton, Southern Stillwater Range (IXL)							
398	39.630	−118.181	5/8	43	27	27.4	9	337‡	41†				
399	39.630	−118.183	7/8	55	46	7.4	68	308‡	43	308	43	44	149
								330§	45	330	45	62	133
400	39.630	−118.183	8/10	30	60	4.5	153	301‡	24	301	24	32	141
								316§	28	316	28	44	132
401	39.630	−118.185	7/10	43	55	12.4	25	303‡	32	303	32	36	145
								321§	36	321	36	51	133
402	39.630	−118.189	4/9	235	−32	10.3	81	146‡	−50	146	−50	−61	325
								170§	−47	170	−47	−76	281
403	39.656	−118.216	9/9	238	−27	5.0	109	152‡	−54	152	−54	−67	328
								178§	−49	178	−49	−80	252
404	39.656	−118.215	6/8	240	−42	14.4	23	130‡	−49	130	−49	−48	334
								155§	−50	155	−50	−68	316
405	39.656	−118.213	4/6	24	10	26.4	13	356‡	23†				
406	39.651	−118.203	9/9	64	−54	25.0	5	75‡	20†				
407	39.651	−118.200	9/9	53	45	15.0	13	310‡	43	310	43	46	148
								332§	44	332	44	63	130
408	39.652	−118.195	6/9	41	25	22.9	10	340‡	40	340	40	66	113
								357§	34	357	34	69	70
409	39.653	−118.193	7/9	119	−45	23.5	8	114‡	28†				
620	39.612	−118.269	7/7	73	59	3.7	265	307	41	313	66	56	182
621	39.611	−188.260	7/7	79	5	3.7	273	26	60				
622	39.598	−118.293	5/7	107	39	4.5	292	267	76				
623	39.598	−118.293	7/7	79	43	3.1	380	286	60				
624	39.620	−118.299	8/8	68	78	5.0	123	287	46	261	49	12	180
625	39.598	−118.302	7/7	101	54	2.3	669	234	52				

At each location the units are listed in approximate stratigraphic order (top of list are oldest). Positive latitudes and longitudes are N and E for locations of sites and virtual geomagnetic poles (VGPs). N/N_0, number of samples accepted/ number of samples demagnetized. D_0/I_0 and D_1/I_1, in situ and tilt-corrected site mean declination and inclination, respectively. D_2/I_2, tilt-corrected declination and inclination of cooling unit mean. a_{95}, cone of confidence about mean direction. k, precision parameter [*Fisher*, 1953].

*Cooling unit means included in composite sequence of units from central part of the study area.

†Site means omitted in calculation of unit mean.

‡For site means from the IXL pluton of the Southern Stillwater locality, the direction is corrected for a 182°/78°W attitude.

§For site means from the IXL pluton of the Southern Stillwater locality, the direction is corrected for a 193°/67°W attitude.

The full ash-flow tuff sequence at the White Rock Canyon locality recorded at least six changes of field polarity. The multiple polarity states recorded by the units, the presence of angular unconformities and overlapped faults within the ash-flow sequence, and the limited K-Ar ages available for the tuffs indicate that a long period of time is represented by the sequence (several million years). Both the mafic sills and the capping basalt recorded normal-polarity directions, similar to the youngest ash-flow tuffs. A time-averaged locality mean derived from the 15 unit means has a direction of D = 327°, I = 61° (Figure 3*a*, Table 2).

Western Stillwater Range. The geologic setting of tuffs sampled in the western Stillwater Range locality (WS, Figure 1) is not well constrained. No detailed mapping of the ash-flow tuff sequence exists for the area. Ten cooling units were sampled in the ash-flow tuff sequence. The base of the sequence was not exposed where sampling was conducted. The tuffs from this locality are all thin and appear to be outflow units. The New Pass Tuff is the uppermost unit of the sequence. Two diabase sills that intrude poorly welded, biotite-rich ash-flow tuffs also were sampled for paleomagnetic analysis. Older ash-flow tuffs at this locality appeared to dip more steeply eastward (as great as 43°) than younger tuffs (with dips as small as 17°), although this relation is complicated by faulting within the sampling area.

A total of 18 sites from 12 units were analyzed from this locality (Figure 3*b*). At least four changes of polarity were recorded by the ash-flow tuff sequence. The two diabase sills have normal magnetic polarities. A locality mean calculated from 12 tilt-corrected unit means is D = 327°, I = 58° (Figure 3*b*, Table 2).

Central Clan Alpine Range. A prominent butte in the central Clan Alpine Range (CCA, Figure 1) exposes numerous ash-flow tuff units. No angular unconformities were observed between units of the sequence. The sequence dips about 12° east.

A total of 17 sites from 12 units were collected at the locality. At least eight field reversals were recorded by the ash-flow tuff sequence. The mean direction of 12 tilt-corrected unit means is D = 329°, I = 63° (Figure 3*c*, Table 2).

Northern Clan Alpine Range. At the northern Clan Alpine locality (NCA, Figure 1) a sequence of relatively thin outflow ash-flow tuff units dip about 30° east. No angular unconformities where observed between units of the sequence. Samples were collected from six sites from five units at this locality. Minimally, four field reversals were recorded by the sequence. The mean direction of five tilt-corrected unit means is D = 336°, I = 59° (Figure 3*d*, Table 2).

New Pass. Sites from this locality (NP, Figure 1) were collected where Highway 50 crosses the southern New Pass Range just west of New Pass Summit. Three discrete ash-flow tuff units were sampled at this locality. The youngest tuff is the New Pass Tuff [*McKee and Stewart*, 1971].

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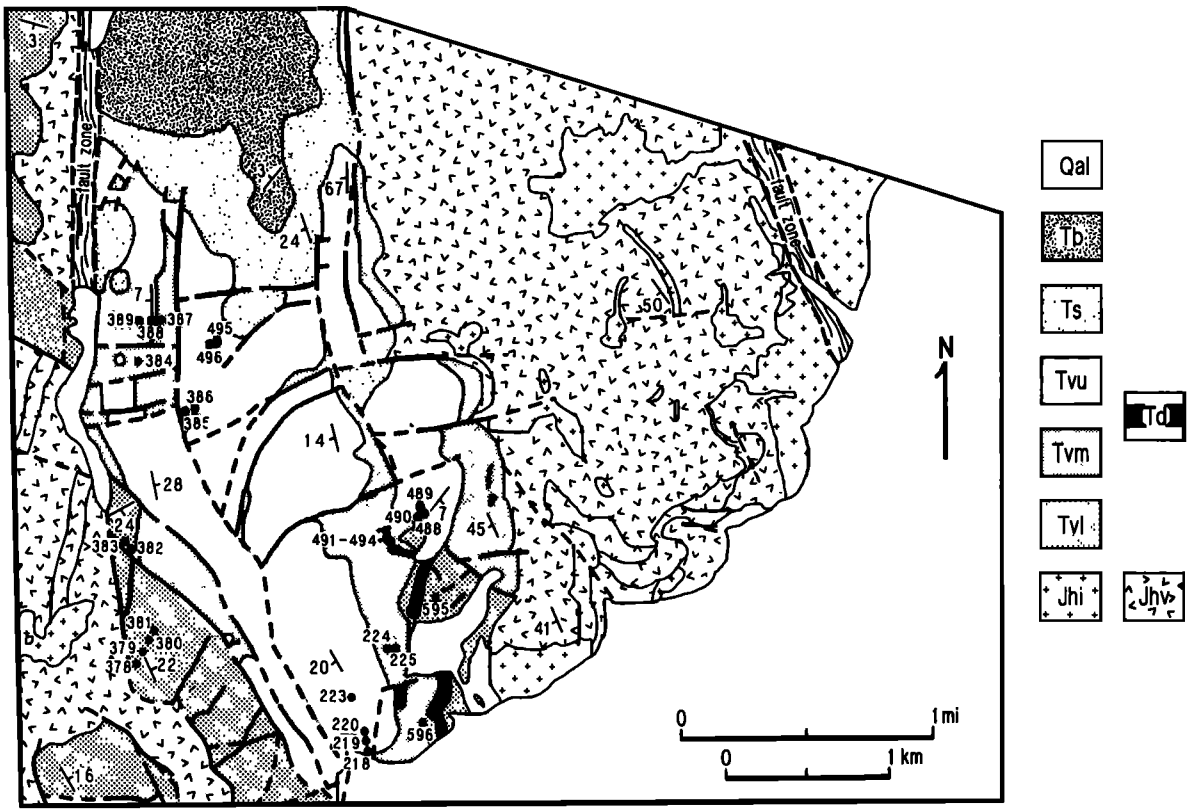


Fig. 4. Generalized geologic map of the White Rock Canyon area of the Stillwater Range [from *Hudson*, 1988] showing locations of paleomagnetic sampling sites from the locality. Qal, Quaternary alluvium and landslide deposits; Tb, middle Miocene basaltic rocks; Ts, Miocene sedimentary rocks; Tvu, Tvm, and Tvl, upper, middle, and lower parts, respectively, of Oligocene to lower Miocene ash-flow tuff sequence; Td, Tertiary diabase sills and dikes; Jhi and Jhv, Jurassic intrusive and volcanic rocks, respectively.

Underlying the New Pass Tuff is a pumice-rich tuff that was also found at several other localities. An older poorly welded tuff and a bedded tuff sequence lying between the New Pass Tuff and the pumice-rich tuff were also sampled. The units are offset by several normal faults, but they are nearly flat lying.

Samples were collected from several sites in the New Pass Tuff where it exhibited different degrees of welding, but no systematic differences in direction of remanence related to degree of welding were observed among the different sites. Data from different horizons of an approximately 5-m-thick section of bedded tuff gave similar directions and were similarly dispersed, so these data were combined to derive a single unit mean.

TABLE 2. Oligocene to Early Miocene Rocks, Locality Means and Rotation Estimates

Locality	*N*	*D*	*I*	a_{95}	*k*	ASD	*R*	*F*
WRC	15	327	61	11.6	12	35	−28 ± 20	−2 ± 9
WS	12	327	58	11.3	15	32	−28 ± 18	1 ± 9
CCA	12	329	63	12.9	12	32	−26 ± 23	−4 ± 10
NCA	5	336	59	18.7	18	31	−19 ± 30	0 ± 15
NP	4	358	69	15.2	37	25	3 ± 37	−10 ± 12
GC	6	349	53	23.3	9	35	−6 ± 32	5 ± 18
EGSD	8	345	38	16.5	12	19	−10 ± 17	21 ± 13
SS	10	310*	47	11.3	19	23	−45 ± 14	12 ± 9
	10	329†	48	13.2	14	30	−26 ± 16	12 ± 10
Composite	25	332	62	8.7	12	35	−23 ± 15	−3 ± 7
Lower	13	318	64	11.3	14	34	−37 ± 21	−5 ± 9
Upper	12	344	58	13.8	11	34	−11 ± 22	1 ± 11

WRC, White Rock Canyon; WS, western Stillwater; CCA, central Clan Alpine; NCA, northern Clan Alpine; NP, New Pass; GC, Golconda Canyon; EGSD, Eastgate–southern Desatoya; SS, southern Stillwater Range. *N*, number of means used in calculation. *D* and *I*, declination and inclination of locality mean. a_{95}, 95% statistical confidence cone. *k*, precision parameter [*Fisher*, 1953]. ASD, angular standard deviation of virtual geomagnetic poles. *R* and *F*, rotation and flattening parameters (*Beck* [1980] with error limits after *Demerest* [1983]).

*,†For the southern Stillwater locality, locality means derived after tilt corrections of N2°E/78°W and N13°E/67°W, respectively, were applied to data from the IXL pluton.

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Nine sites were collected at the locality. Only normal-polarity field directions were recorded by the units. When the four unit directions are combined, the average direction is $D = 358°$, $I = 69°$ (Figure 3*e*, Table 2).

Golconda Canyon, Tobin Range. The northernmost locality of the study is located in Golconda Canyon in the Tobin Range (GC, Figure 1). The geology of the canyon and the area to the south was mapped by *Burke* [1977]. Within the area, Burke identified the oldest ash-flow tuff as the Caetano Tuff, which has been dated as about 33 Ma [*Naeser and McKee*, 1970]. Two cooling units were sampled in this tuff, as well as a thin overlying tuff and an andesite flow. *Burke and McKee* [1979] refer to ages of rhyolites within the andesite sequence overlying the Caetano Tuff as being the same or only slightly younger than the Caetano Tuff. Most sampling concentrated on a well-layered sequence of basalt flows beneath the Caetano Tuff at the head of the canyon on the western range flank. We assume that the basalt flows are related to development of the northern volcano–tectonic trough of *Burke and McKee* [1979] and are Tertiary in age. Twelve sites from separate flows were collected in this sequence. The basalts dip about 20° southeast.

Low dispersion among the site means from the basalt flows (angular standard deviation of VGPs = 5°) indicates that the basalts were erupted and cooled rapidly with respect to secular variation of the geomagnetic field (Figure 3*f*). The 12 site mean directions recorded by the basalts probably represent a single independent short-lived field direction. Therefore of the 19 sites sampled in Golconda Canyon, only 6 independent field directions from three magnetic-polarity intervals are used to estimate the locality mean, $D = 349°$, $I = 53°$ (Figure 3*f*, Table 2).

Augusta Range. The 25-Ma Fish Creek Mountains Tuff [*McKee*, 1970] and a series of andesites, possibly correlative with the Oligocene andesites in the Tobin Range, were sampled at the western margin of the Augusta Range (AR, Figure 1). The margin of the Fish Creek Mountains caldera [*McKee*, 1970] lies about 4 km northeast of the locality. The andesites are predominantly thin flows with irregular boundaries intercalated in volcanogenic sedimentary rocks. Older flows appear to dip more steeply northeast (45°) than younger flows (30°) about a west-northwest strike. Samples were collected at sites in 12 individual flows. The overlying Fish Creek Mountains Tuff has a northwest strike and low southwest dip (6°) that may be primary, given its proximity to the caldera margin. Samples were collected from four sites in different parts of the ash-flow sheet.

Thermal and AF demagnetization isolated tightly grouped magnetizations ($k > 50$) from both andesite flows and tuff. Before tilt correction, site mean directions from the andesite flows exhibit little directional dispersion. They have northwest declination with a moderate positive inclination except for two sites (Figure 3*g*). Interbedding of the flows and sediments and an apparent differential tilt between older and younger flows suggest that significant time lapsed during andesite eruption, so low dispersion among the andesite site mean directions probably cannot be explained by a rapid eruption rate with respect to secular variation of the geomagnetic field. We propose that magnetization in the andesites was thermochemically reset during later development of the Fish Creek Mountains caldera. Of four sites from the overlying Fish Creek Mountains Tuff, three have directions statistically identical to those recorded by most andesite flows (Figure 3*g*). A similar direction from two sites from an intracaldera part of the Fish Creek Mountains Tuff ($D = 344°$, $I = 38°$) was reported by *McKee* [1970]. The third site, from the basal part of the tuff, has a more northwesterly declination. Of the two andesite flows that apparently escaped remagnetization, one has a northerly direction with steep inclination ($D = 350°$, $I = 60°$) and the other records an abnormal direction ($D = 340°$, $I = -30°$) that probably records a transitional field. How these two sites escaped remagnetization is uncertain. The mechanism of remagnetization may have been a selective thermochemical process rather than a purely thermal overprint. With only two determinations of typical field directions, no time-averaged mean direction could be calculated for this locality.

Eastgate and southern Desatoya ranges. Samples were collected along a traverse along Highway 2 extending from the Eastgate Range to Carroll Summit in the southern Desatoya Range (EGSD, Figure 1). This traverse lies in the southern volcano–tectonic trough of *Burke and McKee* [1979], and therefore some tuffs were sampled in probable intracaldera settings. Structural and stratigraphic relations among units sampled are poorly known. Most mapping of this area is reconnaissance in nature. *McKee and Conrad* [1987] published a generalized map of part of the Desatoya Range along and north of Highway 2 after our sampling was conducted. *McKee and Stewart* [1971] and *McKee and Conrad* [1987] have reported K-Ar ages for tuffs and lavas from this area of 24–26 Ma. We sampled moderately to densely welded ash-flow tuffs as well as one air-fall tuff (total of nine sites). Two sites were located in areas mapped by *McKee and Conrad* [1987] as andesite, and three sites were located in the rhyolite of Campbell Creek. Structural attitudes of the tuffs were highly variable along the traverse. Most sites had north strikes and dipped from 12° to 65° west, but a few sites had northeast strikes and dipped southeast as much as 75°.

Demagnetization response of these rocks was variable. At four sites, well-grouped discrete components of magnetization which are probably thermoremanent magnetizations (TRM) were easily isolated using AF demagnetization (Figure 5*a*). At five sites, AF demagnetization was not effective for isolating a discrete component of magnetization, as shown by curvilinear demagnetization paths on vector diagrams (Figure 5*b*) and dispersed directions of magnetization. Thermal demagnetization was more effective for samples from these sites (Figure 5*c*). This demagnetization behavior suggests that coercivity spectra of a secondary chemical remanent magnetization may overlap the coercivity spectra of a primary TRM in these sites, although, unblocking temperature spectra of the different components apparently do not overlap. Secondary alteration may have formed maghemite, as indicated by thermomagnetic curves (Figure 6) from some samples. For samples containing multicomponent magnetization, the component typically removed from 400° to 600°C was used for calculation of the site mean.

Given the uncertainties regarding the geologic setting and the complex character of magnetization in some of the rocks from the traverse, it is encouraging that the tilt-corrected site mean directions are similar to expected Tertiary field directions. Two in situ site means have northerly declinations with negative inclinations (Table 1). However, after tilt correction, all site means have either northerly declinations with positive inclinations or southerly declinations with

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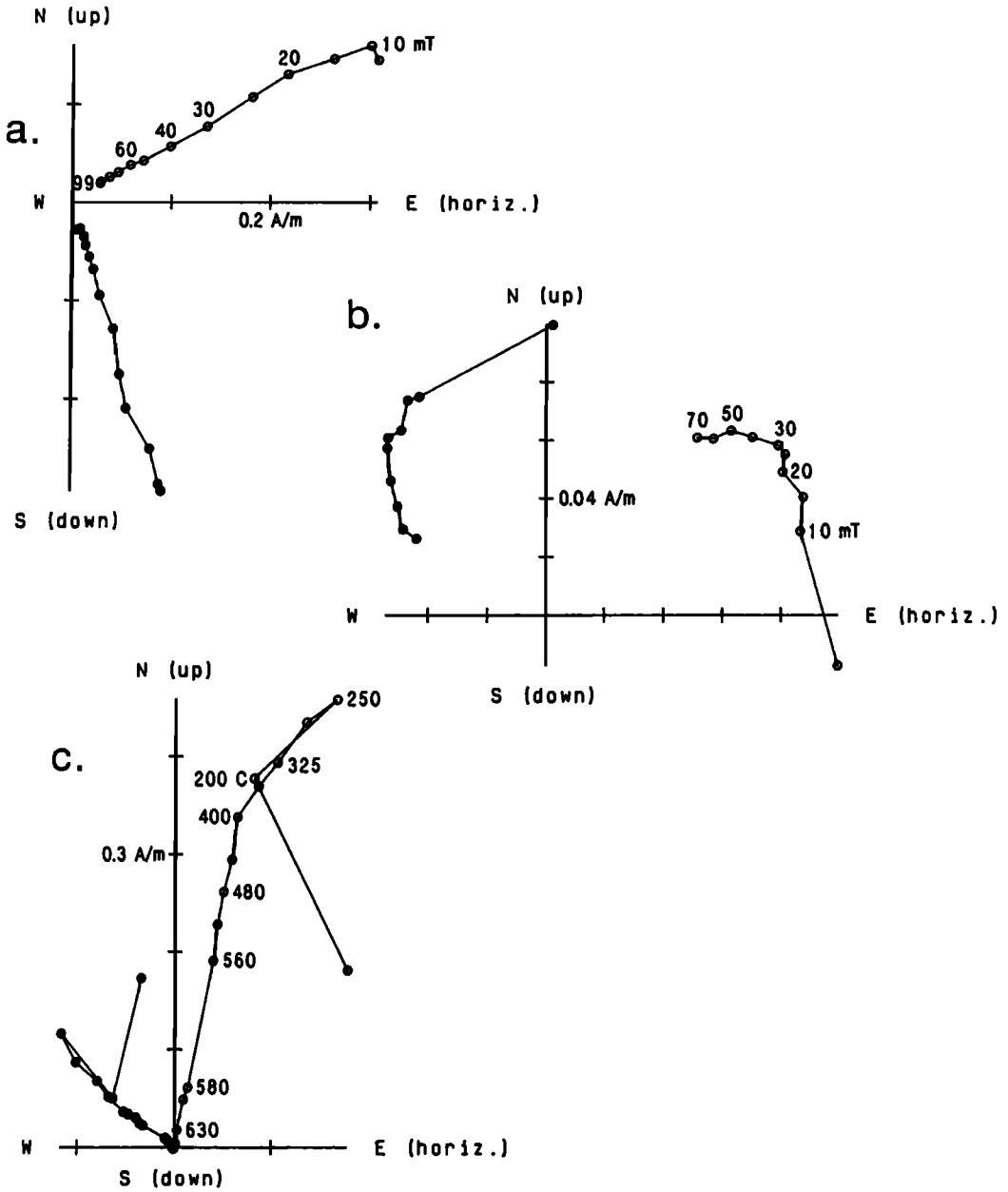


Fig. 5. Orthogonal demagnetization diagrams, shown in in situ coordinates, from Oligocene to lower Miocene volcanic rocks from the Eastgate–southern Desatoya Range locality. Solid and open circles are projections on horizontal and vertical planes, respectively. Demagnetization levels are given in milliTesla (mT) or degrees Celsius (C). (*a*) Ash-flow tuff, site 359, that displayed straightforward behavior during AF demagnetization. (*b*) Densely welded ash-flow tuff, site 354, containing multiple components of remanent magnetization that were not well isolated by AF demagnetization. (*c*) Densely welded ash-flow tuff, site 354, containing a discrete component of remanent magnetization that was isolated above 400°C.

negative inclinations (Figure 3*h*), yielding a positive fold test at the 95% confidence level [*McElhinny*, 1964]. Dual polarities were recorded, but it is uncertain how many polarity epochs were represented by the units sampled. Combining two site means of a tuff collected at Eastgate and assuming that the remaining seven site means record independent field directions, the locality mean is $D = 345°$, $I = 38°$ (Figure 3*h*, Table 2).

Southern Stillwater Range: IXL pluton. Tertiary rocks in the southern Stillwater Range (SS, Figure 1) are the westernmost exposure of the southern volcano–tectonic trough of *Burke and McKee* [1979]. The IXL pluton (Figure 7) is nonfoliated granodiorite to quartz monzodiorite. *Page* [1965] reported a K/Ar age of 29 Ma on hornblende from the pluton which is concordant with a more recent K/Ar determination from biotite (D. A. John and E. H. McKee, unpublished data, 1989). A thick sequence of Tertiary ash-flow tuffs and lavas lie west and south of the pluton. This volcanic sequence probably includes several intracaldera deposits (D. A. John, personal communication, 1987). On its west margin, the IXL pluton partly intrudes a thick tuff with which it may be comagmatic on the basis of geochemical similarities (D. A. John, personal communication, 1989). Foliations in the tuff consistently dip steeply west. Younger

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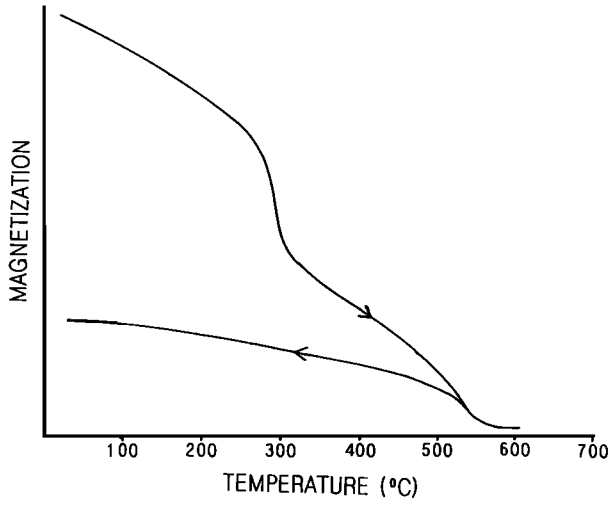


Fig. 6. Thermomagnetic curve from ash-flow tuff sample from site 354, Eastgate–southern Desatoya locality. The step-like loss of magnetization at about 300°C during heating indicates the presence of maghemite in the sample in addition to a probable magnetite phase with a Curie temperature of about 570°C. The sample was magnetized in an 0.18-T induction and heated and cooled in air.

tuffs that have K/Ar ages of about 24 Ma (D. A. John and E. H. McKee, unpublished data, 1989) lie farther to the west and also have steep west dipping foliations. Samples for paleomagnetic analysis were collected from 12 sites along two traverses in the IXL pluton. Samples were also collected from two 24-Ma tuffs at six sites west of the pluton.

For the IXL pluton, results of pilot demagnetization indicated that thermal demagnetization was more effective than AF demagnetization in isolating components of magnetization. Most samples were thermally demagnetized. After removal of a VRM by 400°C, samples from some sites had simple, single-component magnetizations unblocked by 600°C (Figure 8*a*). At other sites, demagnetization behavior above 400°C was more complicated, and vector diagrams show that more than one component of magnetization was unblocked over temperatures of 400°–600°C. Samples that displayed this behavior commonly had a very distinct magnetization component (Figure 8*b*) removed at the highest unblocking temperatures (typically 560°–590°C). The high unblocking temperature component was used for calculation of site means. Dual-polarity components of magnetization were isolated in the IXL pluton. In situ, these components generally have a northeast declination with positive inclination or a southwest declination with negative inclination (Figure 9*a*). Demagnetization response, Curie temperature determinations, and IRM acquisition data all indicate that magnetite is the predominant carrier of magnetization in the pluton. In particular, small (<5 μm) magnetite grains within feldspar grains that are likely carriers of the discrete high unblocking temperature magnetization were observed. The high unblocking temperature components are interpreted to be TRMs.

Dispersion of remanent magnetization in the pluton probably reflects secular variation of the geomagnetic field, recorded as sites slowly cooled. Within-site dispersion is relatively high (k are typically <20, Table 1) for high unblocking temperature components in the pluton (Figure 9*a*). The presence of dual-magnetic-polarity components indicate that cooling of the pluton must have spanned at least one geomagnetic field reversal. Whereas most characteristic magnetizations measured in samples from the pluton have either northeast-down or southwest-up directions, many lie outside this concentration. These magnetizations may have been acquired during transitional field positions or they may reflect incomplete separation of magnetization components of dual polarity in a sample.

An in situ mean direction for the pluton (D = 50°, I = 42°, a_{95} = 10°, n = 8 sites) was derived by omitting four site means with a_{95} > 25° and omitting sample directions highly discordant from the site mean direction for the remaining sites. The means of the accepted normal and reverse polarity sites, however, are not antipodal (the arc distance between the mean of the normal direction and inverted reverse direction is 16°). The shift between of directions of the normal- and reverse-polarity means could reflect incomplete removal of a younger viscous component with a north declination and moderate positive inclination. Averaging both the normal- and reverse-polarity site means should largely cancel the bias of such a younger overprint [*McElhinny*, 1973, p. 85].

AF demagnetization of samples from the two tuffs isolated a high coercivity remanent magnetization that is well grouped at the site level. There is considerable dispersion between site mean directions from each unit, even after tilt correction. This dispersion could reflect secular variation recorded during extended cooling of these thick units or it might arise from errors in tilt corrections.

Locality means were estimated for the southern Stillwater Range by combining tilt-corrected site means from the pluton with tilt-corrected unit means for the two tuffs. Two possible tilt corrections were applied to data from the pluton. An attitude (N2°E/78°W) calculated as the average of 161 foliation measurements in the tuff intruded by and thought to be comagmatic with the IXL pluton was used for one tilt correction (foliation data from D. A. John, unpublished mapping, 1989). With data from the pluton combined with unit means from the tuffs, the locality mean is D = 310°, I = 47° (Figure 9*b*). A second attitude (N13°E/67°W) used for tilt correction for the pluton site means was calculated as an average of 88 foliation measurements from the older of the two sampled 24-Ma tuffs that lie to the west of the pluton. Combining tilt-corrected means for the pluton and the tuffs in this manner yields a locality mean of D = 329°, I = 48° (Figure 9*c*). Using attitudes from tuff units both older and younger than the pluton should bracket the range of possible tilt corrections for the pluton, but both tilt corrections assume that pluton behaved as a rigid body and that there was no difference in tilt between the areas containing the tuffs and the pluton.

Correlations of Ash-Flow Units and Composite Section

A tentative composite magnetostratigraphy of the outflow ash-flow sheets from the central part of the study area was compiled to permit calculation of a more tightly constrained time-averaged mean direction from the paleomagnetic data. Parts of the ash-flow sequences at the different localities are correlative, but the same cooling units are not present at each locality. In compiling the composite magnetostratigraphy, the goal was to include and separate all cooling units that recorded independent field directions without repeating the same unit from different localities. Figure 10 shows, in best estimate of stratigraphic order, the site mean directions

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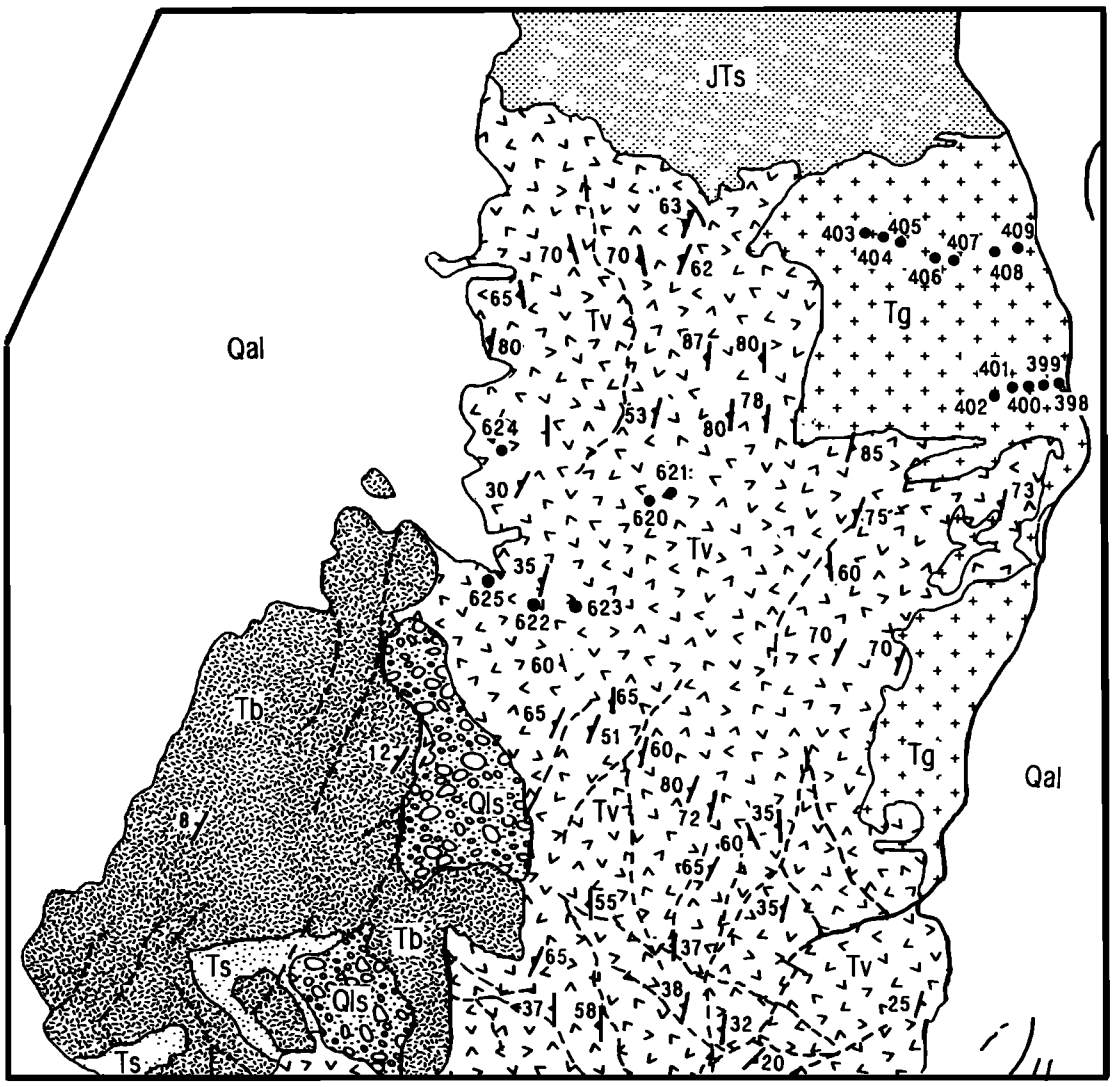


Fig. 7. Generalized geologic map of the southern Stillwater Range [from *Page*, 1965], with locations of paleomagnetic sampling sites from the locality. Qal, Quaternary alluvium; Qls, Quaternary landslide deposits; Tb, Miocene basaltic rocks; Ts, Miocene sedimentary rocks; Tv, Oligocene to lower Miocene volcanic rocks (principally welded ash-flow tuffs); Tg, Tertiary granitic pluton (northern mass is 29-Ma IXL pluton); JTs, Jurassic and Triassic metasedimentary rocks.

of each unit sampled at the New Pass, northern Clan Alpine, central Clan Alpine, White Rock Canyon, and western Stillwater localities. These sequences are correlated on the basis of a distinctive pumice-rich tuff present at the five localities. This tuff may be the 24-Ma unit *D* of the Bates Mountain Tuff (C. S. Gromme, personal communication, 1988). Reliable lithologic correlations can also be made for the 23-Ma New Pass Tuff that directly overlies the pumice-rich tuff and a series of biotite-rich tuffs lower in the sequence that may be correlative with the 27-Ma tuff of McCoy Mine of *McKee and Stewart* [1971].

The stratigraphic sequence at White Rock Canyon was used as a reference to which correlations of tuffs from other localities were made. Once correlations of the New Pass Tuff and pumice-rich and biotite-rich tuffs were made, intervening units with similar magnetizations and lithology were tentatively correlated. Units that did not correlate with the White Rock Canyon sequence were added to the composite stratigraphic section. The composite section may underestimate the true number of individual cooling units recording individual field directions in the central part of the study area because many units tentatively correlated between localities on the basis of similar remanence directions may be temporally distinct. Unit means from the diabase sills and basalt flows from the White Rock Canyon and western Stillwater localities were also included in the compilation. These rocks were assigned a stratigraphic position broadly equivalent to the upper parts of the ash-flow tuff sequence. The composite section comprises a total of 25 individual units. The mean direction of the 25 units is $D = 325°$, $I = 62°$ (Figure 11, Table 2). This mean direction is statistically indistinguishable from the individual locality means from the Stillwater and Clan Alpine Ranges.

The tentative correlations made from the composite magnetostratigraphy (Figure 10) indicate that the White Rock Canyon and central Clan Alpine localities contain units spanning the entire composite section. The western Stillwater and northern Clan Alpine localities contain the upper half

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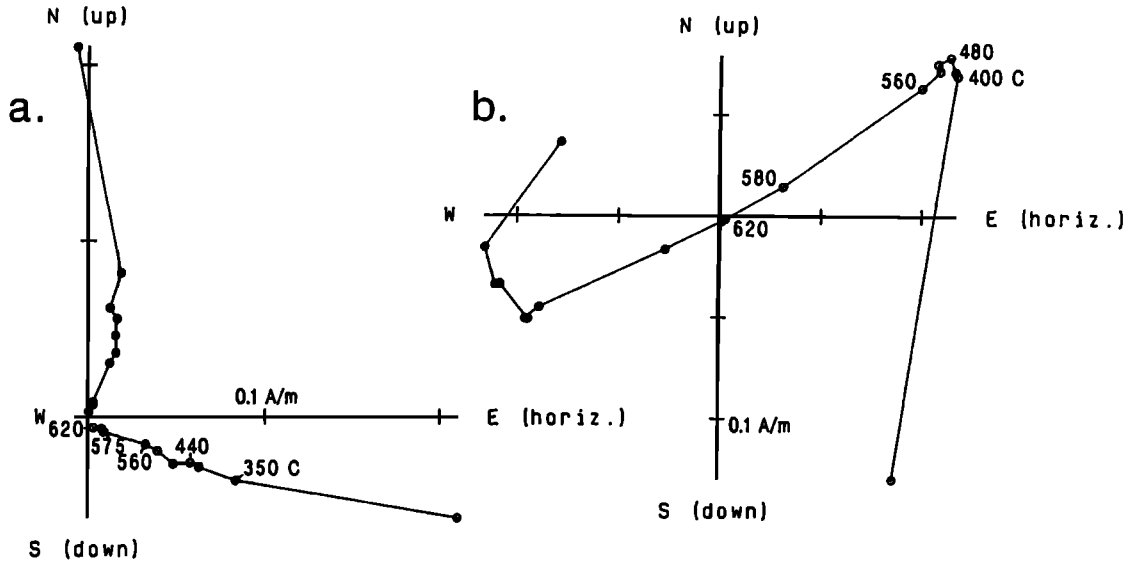


Fig. 8. Orthogonal demagnetization diagrams, shown in in situ coordinates, for samples from the Oligocene IXL pluton, southern Stillwater Range locality. Solid and open circles are projections on horizontal and vertical planes, respectively. Demagnetization levels are given in degrees Celsius (C). (*a*) Sample from site 398. (*b*) Sample from site 403.

of the composite sections, whereas only the uppermost units were sampled at the New Pass locality.

Rotation Estimates

Rotation from time-averaged means. Estimates of vertical-axis rotation (*R* values, Figure 12) for the Oligocene to lower Miocene rocks are of similar magnitude and statistically significant for the White Rock Canyon, western Stillwater, southern Stillwater, and central Clan Alpine locality means ($R = -45°$ to $-26°$) and the composite mean ($R = -23°$) for the central localities of the study area (Table 2). The *R* value for the northern Clan Alpine locality mean ($-19°$) is of similar sense and magnitude to *R* values cited above, but it does not exceed its confidence limits that are large in part because of the small number of independent unit means sampled at his locality. As reported by *Hudson and Geissman* [1987], these data suggest that rocks from the central part of the study area have been rotated counterclockwise. Rotation estimates for means for the northernmost Golconda Canyon locality, the New Pass locality, and the southernmost Eastgate–southern Desatoya traverse are of lesser magnitude ($\leq 10°$) and are not statistically significant. This suggests that rocks sampled at these three localities did not experience significant vertical-axis rotation, although statistical limits for these estimates are large.

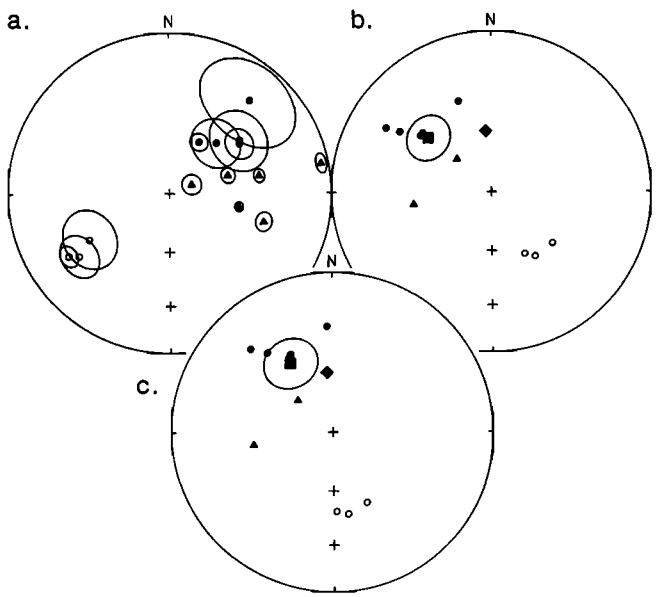


Fig. 9. Equal-area projections of paleomagnetic data from the southern Stillwater locality. Solid and open symbols are projections on lower and upper hemispheres, respectively. (*a*) Accepted site means from the IXL pluton (circles) and Tertiary tuffs (triangles) in in situ coordinates. (*b*) Expected time-averaged direction (diamond) and locality mean (square with 95% confidence cone) derived from tilt-corrected units means from the tuffs (triangles) and the pluton (circles). Pluton data corrected for N2°E/78°W tilt derived from foliations in an intruded tuff. (*c*) As in Figure 9*b* but pluton data corrected for N13°E/67°W tilt derived from foliations in a younger 24-Ma tuff.

The *R* values derived from paleomagnetic data may be inaccurate if tilt corrections were improper or if secular variation was not adequately averaged by the data. We do not believe such errors are generally significant in our data for reasons discussed below.

Simple rotation about a horizontal axis was used for all of the tilt corrections. As discussed by *MacDonald* [1980], declination errors may arise if this simple tilt correction is applied to rocks rotated about an inclined axis or rotated about multiple horizontal axes of different trends. We do not believe our locality means have significant declination errors due to inaccurate tilt corrections because, with the exception of the southern Stillwater Range locality and some sites in the Eastgate–southern Desatoya traverse, few of the dips used for our tilt corrections are greater than 30°. Maximum declination errors are small for these moderate dips [*Scott*, 1984]. Also, because the rocks sampled at the localities have both east and west dips, declination errors of both clockwise and counterclockwise sense would be predicted if rocks where rotated about a common inclined axis. No correlation is found between the amount or direction of dip used in tilt

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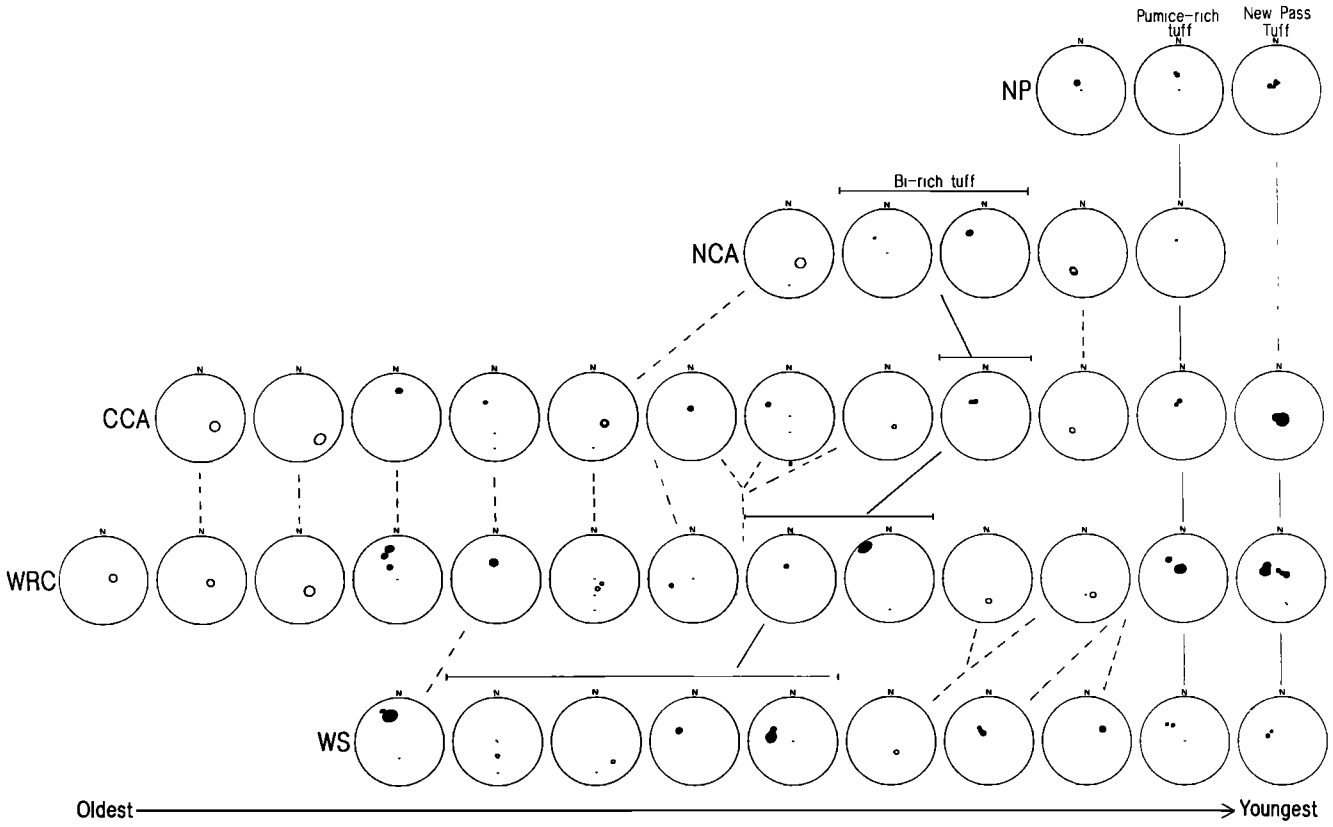


Fig. 10. Equal-area projections of cones of 95% confidence about tilt-corrected site means derived from individual ash-flow tuff cooling units from five localities in the central part of the study area. Solid and open cones are projections on lower and upper hemispheres, respectively. For each locality the data are arranged in best estimated stratigraphic order of units from oldest to youngest on left to right sides, respectively. Correlations of tuffs among the localities are considered reliable for the New Pass Tuff, the pumice-rich tuff, and the biotite-rich tuffs. Other correlations are tentative. NP, New Pass locality; NCA, northern Clan Alpine locality; CCA, central Clan Alpine locality; WRC, White Rock Canyon locality; WS, western Stillwater locality.

corrections and the sense or magnitude of the discordance of unit means from the expected direction.

Inaccurate estimates of the attitude of paleohorizontal also could introduce errors into tilt corrections. Tilt correction for primary dip of strata could produce artificial dispersion of paleomagnetic data. We do not have sufficient information to separate possible primary dip from tectonic dip at many localities. At some localities, such as the White Rock Canyon area, such errors are probably minimal because we principally sampled fault blocks where attitudes were consistent through several successive tuffs. In some instances, attitudes of thin sedimentary horizons intercalated with the tuffs are similar to attitudes of foliations and welding zones in tuffs. Generally, we believe that primary dip of volcanic rocks in most of the area probably is of small magnitude (<10°) and is unlikely to have been systematically oriented. Therefore correction for primary dip is probably not the cause of the statistically significant R values, although it may increase dispersion in the paleomagnetic data.

The uncertainty associated with tilt correction of data from the IXL pluton does warrant caution in interpretation of statistically significant R values for the southern Stillwater locality. For example, if the pluton experienced only 50° of tectonic tilt rather than the 78° or 67° estimated from average foliations of intruded and younger tuffs, the mean for the pluton would parallel the expected direction after tilt correction. However, it seems unlikely that the IXL pluton was less tilted than the younger tuffs to the west. Moreover, declinations of tilt-corrected unit means for both 24-Ma tuffs that lie west of the pluton are counterclockwise from the time-averaged expected direction (Table 1). These observations support the interpretation that the southern Stillwater Range locality also has been affected by counterclockwise vertical-axis rotation.

The accuracy of R values and their confidence limits also depends on whether secular variation has been adequately averaged at each locality. Particular attention was paid to deriving unit means that were independent records of the geomagnetic field so as not to overestimate the statistical significance of locality means. Generally, if it was questionable whether site means from adjacent sites with similar directions represented one or two separate cooling units in ash-flow tuffs, a single unit mean was calculated. Angular standard deviations of VGPs for the locality means (19°–35°, Table 2) that equal or exceed the expected value for this latitude (19°, *Merrill and McElhinny* [1983]) and the record of dual polarities at most of the localities argue that significant secular variation was averaged by most locality means. The greater than expected dispersion of VGPs may reflect errors associated with tilt corrections and (as discussed later) vertical-axis rotation that occurred during the time span of ash-flow deposition.

Flattening (F) values provide another measure of whether the time-averaged means are valid estimates of geomagnetic dipole directions. Given the intracontinental setting, a large

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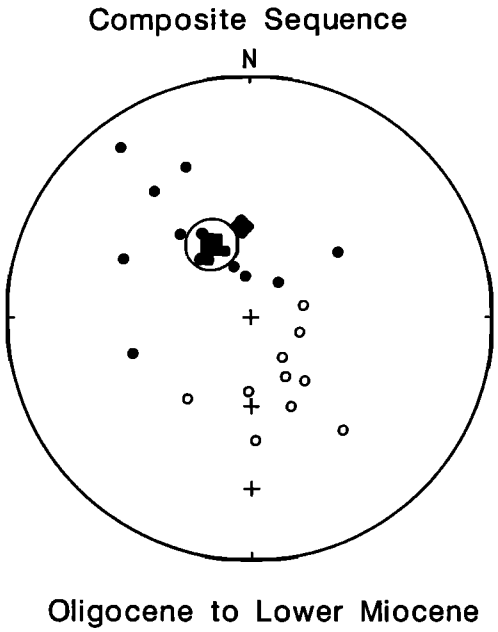


Fig. 11. Equal-area projection of tilt-corrected unit means (circles) from the composite sequence of Oligocene to lower Miocene rocks derived from the White Rock Canyon, western Stillwater, and central Clan Alpine localities from the central part of the study area. Included unit means are identified in Table 1. Also shown are the time-averaged mean (square) derived from the unit means (with cone of 95% confidence) and the time-averaged expected direction (diamond) for the area. Solid and open symbols are projections on lower and upper hemispheres, respectively.

latitudinal translation of the Dixie Valley region since the middle Tertiary would be an unreasonable hypothesis. Therefore no statistically significant flattening values are expected if, within their confidence limits, the time-averaged means are valid estimates of geomagnetic dipole directions. For the time-averaged means from eight localities and the composite sequence listed in Table 2, only two of the means have statistically significant F values. For the southern Stillwater Range locality it seems most likely that errors in tilt corrections are responsible for the F values that slightly exceed their confidence limits (12° ± 9° and 12° ± 10° for the two estimates). The mean for the Eastgate–southern Desatoya locality may inadequately average secular variation considering its angular standard deviation of VGPs (19°) which is the lowest for any of the localities (Table 2) and its statistically significant F value (21° ± 13°). The significant F value also could arise from correction for primary north dip that might have been present along the southern margin of the southern volcano–tectonic trough.

Rotation between localities. For localities containing outflow ash-flow tuff sheets in the central part of the area, differential tectonic rotation among localities can be assessed by comparing directions of magnetization from correlative units. Although the New Pass Tuff was the most extensively sampled unit at four localities, intraunit dispersion and steep inclinations of its remanent magnetization make resolution insufficient for confident detection of vertical-axis rotation. Directions of magnetization from the pumice-rich tuff are more useful. Site mean declinations from the pumice-rich tuff are 334°–354° at the White Rock Canyon, northern and central Clan Alpine, and New Pass localities, but they are 314°–323° at the western Stillwater locality (Figure 10). A declination shift of similar sense and magnitude is seen for magnetization from biotite-rich tuffs of the Stillwater and Clan Alpine Ranges (Figure 10). Because the biotite-rich tuff is composed of several thin cooling-units it is not possible to correlate individual units between localities. However, normal polarity units in biotite-rich tuffs from the White Rock Canyon and Clan Alpine localities typically have declinations of 315°–339°, whereas the biotite-rich tuffs from the western Stillwater locality have declinations of 291–305°. These comparisons suggest that the western Stillwater locality experienced about 20° more counterclockwise rotation than the other localities. There is little evidence for differential rotation among the other localities with correlative ash-flow units.

Evidence for rotation during ash-flow deposition. Declinations of unit mean directions from the composite section, in general, shift counterclockwise with increasing age of units (Figure 13*a*), although this trend is partly obscured by dispersion related to secular variation. The general declination shift is more clearly illustrated by means of successive sets of ten units from the composite sequence (Figure 13*b*) that were calculated using a sliding window analysis. This stratigraphically controlled shift of declinations is unlikely to reflect a coincidental recording of secular variation because of the long period of time spanned, including several reversals, and the large number of units sampled in the composite section. Instead, we infer that rotation was ongoing during ash-flow deposition. The counterclockwise shift of declinations suggests that older units were rotated more than younger units. If the composite section is arbitrarily divided into halves, the rotation estimate from the older 13 units is $R = 37° \pm 21°$ whereas that from the younger 12 units is $R = -11° \pm 16°$. Directions of remanent magnetization in the pumice-rich tuff at all localities, excluding the western Stillwater locality, are similar to directions reported by *Gromme et al.* [1972] for unit D of the Bates Mountain Tuff from sites distributed broadly over central Nevada. If the pumice-rich tuff is correlative with unit D, this would further imply that upper parts of the composite section in the Dixie Valley area have experienced little counterclockwise rotation relative to areas in central Nevada. Rotation values calculated from time-averaged means that incorporate data from a wide age range of tuffs (e.g., the $R = -23°$ for the composite mean) are probably minimum estimates for the total amount of Tertiary rotation because they are in part derived from units younger than much of the rotation.

Vertical-axis rotation that spanned the period of tuff deposition can explain apparent inconsistencies between estimates of rotation of localities from the central part of the area based on the locality means and the unit means from individual ash-flow tuff cooling units. Reviewing, time-averaged locality means suggest that the western Stillwater, White Rock Canyon, central Clan Alpine, and northern Clan Alpine localities were rotated counterclockwise from 19° to 28°, but the New Pass locality was little rotated (Figure 12, Table 2). In contrast, directions obtained from the pumice-rich tuff, one of the youngest units of the composite sequence, indicate little relative rotation occurred between the White Rock Canyon, central Clan Alpine, northern Clan Alpine, and New Pass localities, but about 20° of relative counterclockwise rotation affected the western Stillwater locality (Figure 10). If regional vertical-axis rotation occurred during ash-flow deposition, little or no rotation would be predicted at the New Pass locality because only the

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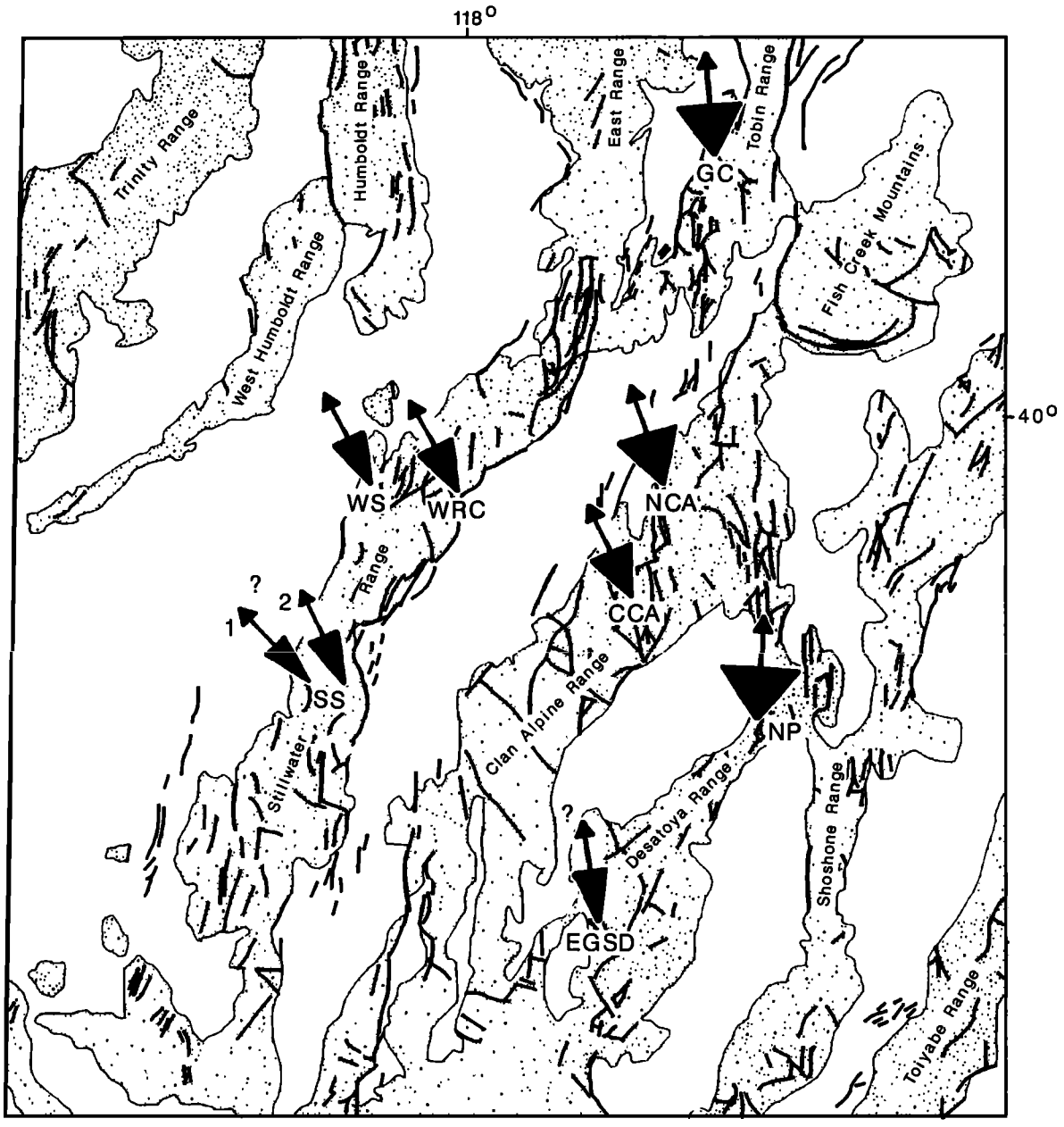


Fig. 12. Map showing estimates of vertical-axis rotation derived from paleomagnetic data from Oligocene to lower Miocene igneous rocks. Rotation magnitude and sense, relative to present geographic north, are depicted by arrows for the localities. Angles subtended by the fans centered on the arrows are 95% confidence limits for the rotation estimates (calculated after *Demarest* [1983]). CCA, central Clan Alpine; EGSD, Eastgate–southern Desatoya; GC, Golconda Canyon; NCA, northern Clan Alpine; NP, New Pass; SS, southern Stillwater; WRC, White Rock Canyon; WS, western Stillwater. For the southern Stillwater locality, different rotation estimates (arrows 1 and 2) reflect different possible tilt corrections applied to data from the IXL pluton (Table 2). Queried arrows represent rotation estimates that are considered less reliable because of uncertainties in tilt corrections or possible inadequate averaging of geomagnetic secular variation, as discussed in the text.

youngest and least-rotated units of the composite stratigraphy were sampled here (Figure 10). Likewise, we infer that rotation was largely finished at the White Rock Canyon, central Clan Alpine, and northern Clan Alpine localities by the time of deposition of the pumice-rich tuff, and therefore declinations of the pumice-rich tuff at the New Pass and these other localities should be similar. Correlations with the composite sequence indicate that only the middle and upper parts of the tuff section were sampled at the western Stillwater locality (Figure 10). Therefore although declinations of the pumice-rich and biotite-rich tuffs suggest that this locality has undergone greater counterclockwise rotation (Figure 10), the rotation calculated from the time-averaged mean for the western Stillwater locality may be similar in magnitude to rotations based on a time-averaged mean from the other central localities (Figure 12, Table 2) because only the younger, less rotated half of the composite sequence was sampled at the western Stillwater locality. This conclusion suggests that older tuffs at the western Stillwater locality, if preserved, should have highly discordant declinations. The deflection of magne-

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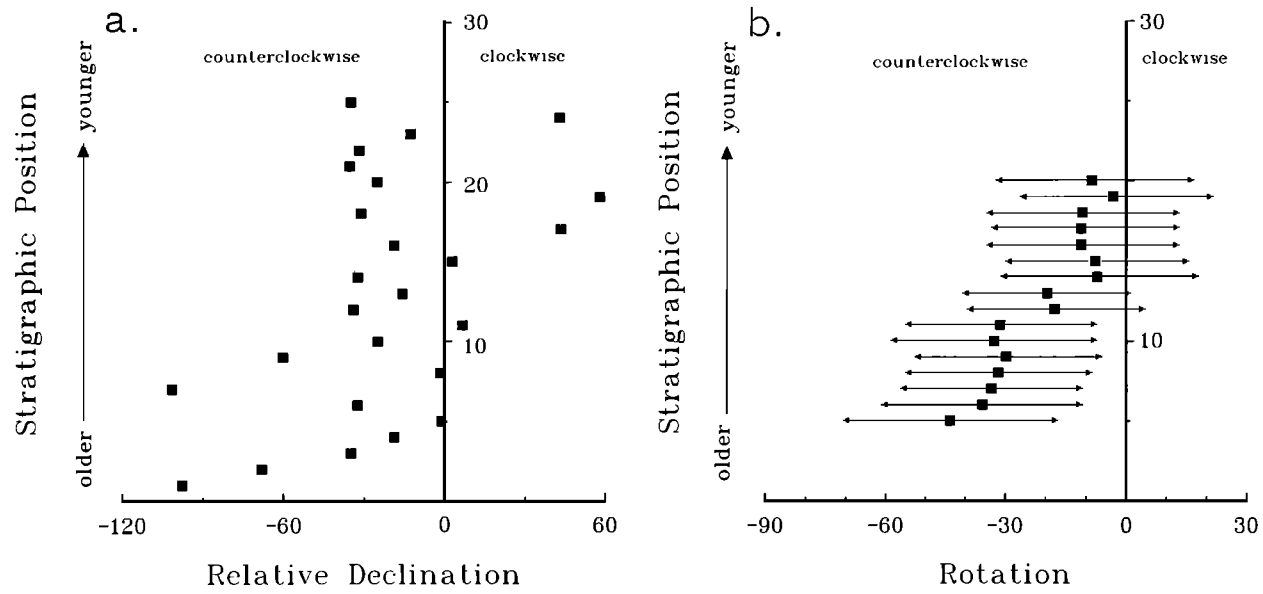


Fig. 13. Evidence for counterclockwise vertical-axis rotation during deposition of the Oligocene to lower Miocene composite sequence from the central part of the Dixie Valley region. (*a*) Declinations of unit means from the composite sequence plotted relative to the declination of a time-averaged expected direction for the area (355°). The declinations show a general counterclockwise shift with increasing age of the units, although this trend is partly obscured by dispersion from secular variation. (*b*) Estimates of vertical-axis rotation (with 95% confidence limits shown by arrows) versus general stratigraphic position for the composite sequence. Rotations are based on a series of means calculated from a sliding window analysis of the sequence. For the analysis, individual windows averaged ten units, and successive windows were moved through the sequence in one-unit increments. The stratigraphically controlled change of declinations, and thus rotations, is more clear because secular variation of the units is largely averaged. The stratigraphically controlled change of declinations is believed to reflect greater counterclockwise rotation of the older units.

tization from the pumice-rich tuff indicates that counterclockwise rotation continued to a younger age at the western Stillwater locality than at the other localities.

Recognition that rotation may have occurred during the time span of ash-flow deposition also influences the interpretation for localities that do not show statistically significant R values. For example, although error limits for the rotation estimate ($R = -6° \pm 32°$) are large for the Golconda Canyon locality, age for all units sampled was at least 33 Ma. Thus if the locality experienced counterclockwise rotation, the age of sampled strata indicate that it should have recorded the maximum possible rotation of any of the localities (≥ the 37° estimated from the older half of the composite sequence). We interpret the Golconda Canyon locality to have escaped Tertiary vertical-axis rotation. In contrast, we suspect that the area of the New Pass locality may have experienced vertical-axis rotation, but, as discussed earlier, the paleomagnetic data from the locality do not reflect this because the units sampled were probably too young to have experienced significant rotation. Rocks sampled at the Eastgate–southern Desatoya locality also may be too young to have experienced significant vertical-axis rotation. The K-Ar ages reported for rocks in the southern Desatoya Range are 24–26 Ma [*McKee and Stewart*, 1971; *McKee and Conrad*, 1987]. However, unlike the New Pass locality, the rocks at some sites from the Eastgate–southern Desatoya locality were highly tilted. Deformation at this area may have been younger than at localities farther north, or the rocks sampled in this study may be older than those sampled for isotopic dating. In either case our paleomagnetic data do not indicate that significant vertical-axis rotation occurred during deformation of this locality. This interpretation is suspect, however, because the locality mean may not have adequately averaged secular variation.

Middle Miocene Basalts

General Characteristics

In contrast to complex stratigraphic and structural characteristics of Oligocene to lower Miocene rocks, geologic relations of the middle Miocene basalts in the study area are relatively simple. The rocks are predominantly thin basalt flows intercalated with smaller volumes of pyroclastic deposits (predominantly agglomerates). Typically, the basalt flows dip less than 10°.

Demagnetization behavior for most of the basaltic rocks was uncomplicated. AF demagnetization typically isolated a well-grouped remanent magnetization in the basalts at peak inductions of 15–90 mT (Figure 14). Lightning induced IRMs were the most common secondary magnetizations in the rocks, although reheating by overlying flows has introduced secondary components in some flows. For most site means, values of k exceed 100 (Table 3). Ranges of unblocking temperatures (200°–600°C) and Curie temperatures (500°–580°C) from representative samples indicate that low-Ti titanomagnetite is the principal carrier of magnetization in most of these basalts. The remanent magnetization in most basalts is probably a simple TRM. Some agglomeratic deposits and flow margins were reddened and contained a significant component of magnetization carried by hematite that formed during early oxidation during initial cooling of the rocks.

Although determination of site mean directions, typically, was simple for the basaltic rocks, determining locality means

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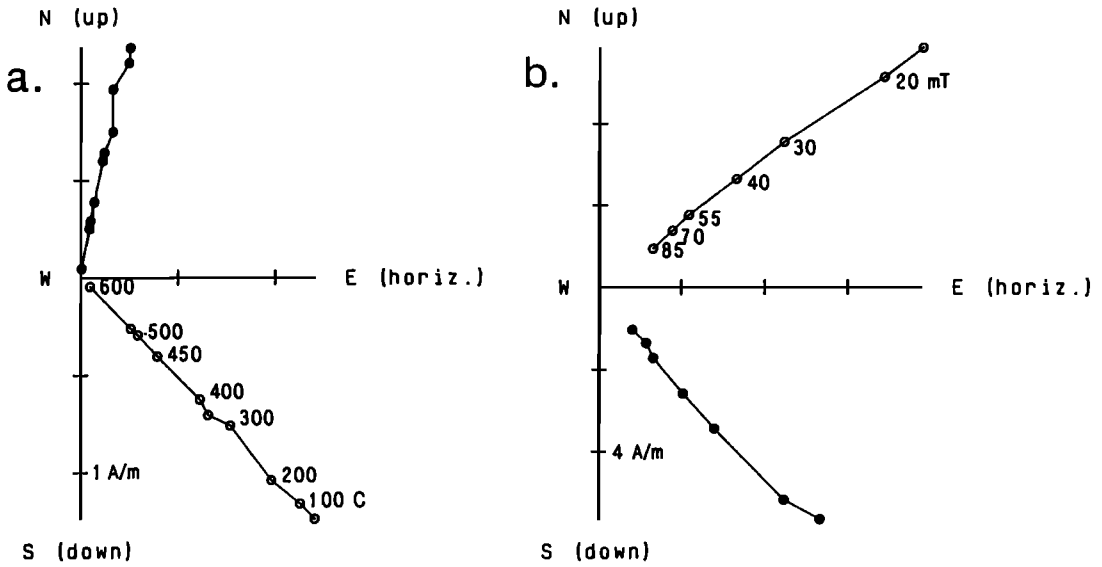


Fig. 14. Representative orthogonal demagnetization diagrams, shown in in situ coordinates, from middle Miocene basaltic rocks from the Dixie Valley region. Solid and open circles are projections on horizontal and vertical planes, respectively. Demagnetization levels are given in milliTesla (mT) or degrees Celsius (C). (*a*) Basalt flow, site 183, Kitten Springs locality. (*b*) Basalt flow, site 289, southeast Humboldt Range locality.

that properly represented time-averaged geomagnetic fields for these rocks was more complicated. Means from groups of adjacent sites often had tightly clustered directions, suggesting that the rocks erupted and cooled rapidly with respect to secular variation. If multiple sites at a locality recorded the same geomagnetic field direction, assigning each site equal weight would bias the time-average mean for the locality toward the overrepresented field direction and would underestimate the dispersion of independent field directions recorded at the locality. This could cause errors in calculation of both R values and their confidence limits for time-averaged means. Similar difficulties were noted by *Calderone et al.* [1990] for paleomagnetic data from Miocene basaltic rocks from western Arizona and southeastern California.

For the middle Miocene basaltic rocks, site mean directions were combined in two manners in an effort to derive independent recordings of geomagnetic field directions. Means from stratigraphically adjacent sites were compared using the method of *McFadden and Lowes* [1981] to determine if they had statistically distinguishable directions. Means from groups of adjacent sites that did not have distinguishable directions were combined to form "unit" means. Each unit mean is believed to record an independent geomagnetic field direction. The unit means were then combined to determine time-averaged locality means. In some cases, however, magnetization from groups of adjacent sites had sufficient dispersion to have statistically distinguishable unit means, but their directions remained clustered with respect to the overall dispersion of site means at the locality. Rock sequences that displayed this clustering of site means may not have been erupted over a sufficient time interval to record more than a limited arc of paleosecular variation. Unit means from such sequences might not be randomly distributed about a mean dipole field direction. Therefore a second more conservative manner of combining the data was to average all unit means within a suspected eruptive sequence and then average these "eruptive period" means to derive a time-averaged mean for a locality. Grouping of sites within individual eruptive periods was somewhat subjective, but typically, arc distances between site means from adjacent sites within a designated eruptive sequence were $<10°$, whereas those between eruptive sequences were $>20°$.

Localities

Southeastern Humboldt Range. Numerous thin (2–3 m), planar middle Miocene basalt flows overlie Triassic metasedimentary rocks at the southeastern flank of the Humboldt Range [*Wallace et al.*, 1969]. Samples from sites in 20 individual stacked flows were collected at this locality (SEH, Figure 1). A thin sediment horizon separates upper and lower parts of the flow sequence. The sequence dips 10° southeast about a north-northeast strike.

Magnetization from sites below the sediment horizon are of normal polarity, whereas those above the horizon are of reverse polarity. Site means from several groups of adjacent flows have statistically indistinguishable directions (Figure 15*a*). Angular standard deviation of site VGPs is 3° for the normal interval and 6° for the reverse interval. These low dispersions indicate that flows within each polarity interval were emplaced in a single eruptive period and cooled rapidly with respect to secular variation of the geomagnetic field. Means of the two polarity intervals are not statistically antipodal. A locality mean based on seven unit means is $D = 327°$, $I = 49°$ (Figure 15*a*, Table 4), but it is unlikely that secular variation is adequately averaged in this mean, considering the limited dispersion among the means.

Muttlebury Canyon, West Humboldt Range. Eight sites were occupied in thin flows along Muttlebury Canyon on the east flank of the West Humboldt Range (MC, Figure 1). Samples were collected from groups of three, two, and three adjacent flows. The groups of flows were stratigraphically separated by several meters. The flows dip southeast 10°.

All site means from the Muttlebury Canyon locality have positive inclinations. From lowest to highest flows, declina-

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TABLE 3. Paleomagnetic Data From Middle Miocene Basalts From the Region Surrounding Dixie Valley, West Central Nevada

	Site											VGP	
Number	Latitude	Longitude	N/N_0	D_0	I_0	a_{95}	k	D_1	I_1	D_2	I_2	Latitude	Longitude
						Southeastern Humboldt Range (SEH)							
280	40.163	−118.095	7/7	335	48	3.3	338	343	56	342	55	75	140
281	40.163	−118.095	8/8	333	46	1.5	1367	340	54				
282	40.164	−118.095	7/7	337	43	4.3	197	344	51	344	51	74	123
283	40.164	−118.095	7/7	336	43	2.7	519	342	51				
284	40.164	−118.095	7/7	339	43	3.3	332	346	50				
285	40.164	−118.095	7/7	338	45	3.6	283	345	53				
286	40.164	−118.095	8/8	141	−45	2.2	649	145	−54	145	−54	−62	332
287	40.164	−118.095	7/7	147	−41	2.5	571	152	−50	152	−50	−65	318
288	40.164	−118.095	5/7	135	−29	3.6	460	137	−39	137	−39	−49	319
289	40.164	−118.095	7/7	135	−36	3.3	327	138	−46	139	−44	−53	322
290	40.164	−118.095	7/7	137	−36	5.1	143	140	−46				
291	40.164	−118.095	7/7	136	−37	4.2	208	138	−47				
292	40.164	−118.095	6/6	138	−37	4.1	267	141	−47				
293	40.164	−118.095	6/6	135	−34	3.8	312	137	−43				
294	40.165	−118.094	8/8	132	−38	2.7	422	134	−47	136	−48	−53	328
295	40.165	−118.094	8/8	130	−39	2.6	457	132	−49				
296	40.165	−118.094	7/7	132	−39	3.9	244	134	−48				
297	40.165	−118.094	8/8	134	−37	4.2	175	136	−47				
298	40.165	−118.094	5/5	134	−37	4.9	246	136	−46				
299*	40.165	−118.094	8/8	137	−41	6.4	···	140	−50				
						Muttlebury Canyon, West Humboldt Range (MC)							
430	40.155	−118.300	6/7	175	59	5.4	153	164	52	164	62	−5	253
431	40.156	−118.298	5/6	192	50	4.6	282	181	46	181	46	−22	241
432	40.157	−118.298	7/7	63	73	3.3	342	83	66	78	66	36	295
433	40.154	−118.295	8/8	51	72	3.0	332	73	67				
435	40.153	−118.294	9/9	67	77	4.6	125	89	69	89	69	31	287
434	40.152	−118.292	8/8	334	40	3.9	207	340	48	340	48	70	124
436	40.151	−118.290	4/7	315	72	6.8	182	331	81	331	81	55	227
437*	40.151	−118.288	7/7	339	61	12.3	···	354	68	354	68	78	223
						Coal Canyon, West Humboldt Range (CC)							
170	40.187	−118.256	5/6	68	73	6.2	155	82	70	91	69	30	287
171	40.187	−118.254	6/7	90	73	5.5	150	99	68				
172*	40.187	−118.254	7/7	35	73	15.5	···	53	72	53	72	52	287
173	40.187	−118.253	5/5	117	45	5.2	215	118	39	118	39	−5	297
174	40.188	−118.252	7/7	106	68	4.1	216	111	63	111	63	15	285
175	40.190	−118.251	3/7	112	54	4.6	705	114	48	114	48	3	295
						Kitten Springs, Stillwater Range (KS)							
183	40.049	−117.913	6/6	14	47	5.8	133	20	48	20	48	71	0
184	40.048	−117.913	4/5	5	38	7.1	168	8	40	11	41	71	30
187	40.048	−117.915	6/6	11	40	2.2	929	15	41				
185	40.047	−117.913	5/7	337	65	3.2	558	344	69	344	69	73	306
186*	40.046	−117.913	7/7	213	−50	9.7	···	219	−50	219	−50	−57	156
176	40.040	−117.926	7/7	139	−45	3.8	259	141	−50	143	−52	−59	329
177	40.041	−117.928	7/7	143	−41	5.9	105	145	−45				
178	40.041	−117.928	6/7	143	−50	7.8	75	145	−54				
179	40.041	−117.928	5/6	139	−56	5.9	247	143	−52				
						Northern Stillwater Range (NS)							
501	40.019	−117.940	7/7	6	34	2.3	704	8	34	8	34	68	42
502	40.019	−117.940	7/7	11	41	3.7	261	14	41	17	40	68	19
503	40.019	−117.940	7/7	13	44	5.3	132	15	43				
504	40.019	−117.940	7/7	15	39	3.0	416	18	39				
505	40.019	−117.940	6/6	17	36	3.3	415	19	36				
506	40.019	−117.941	7/7	3	41	5.6	119	5	41	4	43	74	49
507	40.020	−117.941	6/7	0	44	5.3	163	3	44				
508	40.020	−117.941	8/8	354	38	4.0	189	356	39	356	39	71	74
509	40.020	−117.942	7/7	155	−58	4.2	210	159	−58	157	−58	−72	335
510	40.020	−117.942	7/7	154	−53	2.6	541	157	−55				
511	40.020	−117.942	6/8	150	−60	7.8	75	155	−62				
512	40.020	−117.942	6/8	33	42	6.8	97	35	40	39	45	55	343
514	40.020	−117.943	7/7	38	46	5.8	109	41	45				
515	40.020	−117.943	7/7	39	50	3.6	280	42	48				
513	40.020	−117.943	6/6	28	57	13.1	27	32	56	32	56	65	329
516	40.020	−117.943	7/7	38	41	3.4	317	40	40	40	40	52	347
517	40.020	−117.944	7/7	171	−59	4.4	185	176	−60	175	−58	−86	314
518	40.020	−117.944	7/8	169	−55	3.0	407	174	−56				
519	40.020	−117.944	7/7	161	−66	4.1	217	167	−67	167	−67	−77	23

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TABLE 3. (continued)

Number	Site		N/N_0	D_0	I_0	a_{95}	k	D_1	I_1	D_2	I_2	VGP	
	Latitude	Longitude										Latitude	Longitude
						Northern Stillwater Range (NS) (continued)							
520	40.020	−117.945	6/7	144	−70	2.6	658	151	−72	153	−76	−62	35
521	40.020	−117.945	6/7	145	−78	9.3	53	156	−80				
522	40.020	−117.945	7/7	147	−73	3.3	334	155	−75				
523	40.020	−117.945	6/7	146	−48	1.3	2484	148	−50	148	−50	−62	322
524	40.020	−117.945	7/7	157	−48	5.1	140	159	−50	159	−50	−71	309
525	40.020	−117.946	7/7	150	−55	3.3	343	155	−57	155	−57	−69	334
526	40.020	−117.946	7/7	140	−49	2.7	501	142	−52	142	−52	−59	330
						Sou Hills (SH)							
463	40.144	−117.731	7/7	339	58	2.9	431	339	63	339	63	74	175
464	40.147	−117.729	7/7	198	−56	3.3	329	204	−59	204	−59	−72	145
462	40.146	−117.729	6/7	209	−53	3.9	302	214	−56	214	−56	−63	147
455	40.141	−117.739	7/7	329	32	2.2	731	328	37	328	37	56	126
454	40.141	−117.739	7/7	335	34	2.2	761	335	39	335	39	62	119
453	40.141	−117.739	6/7	324	44	3.1	471	323	49	325	49	60	144
452	40.141	−117.738	7/7	329	44	3.8	250	329	49				
451	40.141	−117.738	7/7	322	36	2.5	604	322	41	322	41	54	136
456	40.143	−117.736	6/7	170	−39	4.9	190	171	−44	171	−44	−74	273
458	40.142	−117.737	5/7	164	−46	2.1	1321	165	−51	165	−51	−75	302
457	40.142	−117.735	5/7	163	−18	3.4	515	163	−23	163	−23	−58	275
466	40.159	−117.722	7/7	352	38	2.6	541	5	51	5	51	81	35
468	40.161	−117.722	7/9	330	25	6.0	103	335	41	339	41	66	115
469	40.160	−117.722	6/7	337	26	6.0	127	343	42				
467	40.159	−117.723	6/8	337	47	4.3	245	351	62	351	62	83	180

At each locality the units are listed in approximate stratigraphic order (top of list are oldest). Positive latitudes and longitudes are N and E for locations of sites and virtual geomagnetic poles (VGPs). N/N_0, number of samples accepted/number of samples demagnetized. D_0/I_0 and D_1/I_1, in situ and tilt-corrected site mean declination and inclination. D_2/I_2, tilt-corrected declination and inclination of cooling unit (defined as site or sequence of sites whose mean is statistically distinguishable from means of sites from underlying and overlying units). a_{95}, 95% confidence cone about mean. k, precision parameter [*Fisher*, 1953].

*Site means determined from intersection of remagnetization circles (reported a_{95} is the average of the major and minor axes of confidence ellipse).

tions of site means change from south to east to northwest (Figure 15*b*, Table 4). The general progression of declinations of these positively inclined magnetizations from south to north suggests that this flow sequence records a change from a transitional to a normal dipole-dominated geomagnetic field. Only the upper three site means with northwest declinations probably reflect normal dipole-dominated geomagnetic field directions. A locality mean is not calculated for the Muttlebury Canyon basalts because of the limited number of site means reflecting dipole field behavior.

Coal Canyon, West Humboldt Range. Six thin flows were sampled in hillside exposures south of the road through Coal Canyon on the east flank of the West Humboldt Range (CC, Figure 1). The flows dip 6° to the southeast.

Most samples from the locality contained lightning-induced IRMs that could be removed during AF demagnetization, although intersection of remagnetization circles was required to estimate the mean for one site. The site means have positive inclinations but atypical east and southeast declinations (Figure 15*c*, Table 4). These flows probably recorded transitional field directions, and they are most likely coeval, in part, with flows carrying easterly declinations in the Muttlebury Canyon sequence 5 km to the southwest. No locality mean was calculated from these transitional field directions.

Kitten Springs Road, Northern Stillwater Range. Samples were collected from nine sites from nearly flat-lying flows and agglomerates in a traverse along a road crossing the Stillwater Range from Kitten Springs on the western flank of the range (KS, Figure 1). Sites from the lower part of the traverse have normal magnetization, whereas sites from the upper part of the traverse have reverse magnetization. Site means from four adjacent reverse flows and two adjacent normal flows have statistically indistinguishable directions and are combined to derive unit means. A locality mean derived from the five unit means is $D = 8°$, $I = 55°$ (Figure 15*d*, Table 4). A locality mean derived from four inferred eruptive period means is $D = 3°$, $I = 57°$ (Figure 15*d*, Table 4).

Northern Stillwater Range. At the northern Stillwater locality (NS, Figure 1), samples were collected from 26 sites in basalt flows and agglomerate horizons in a traverse through a single stratigraphic sequence. Pyroclastic horizons and associated dikes were sampled in the lower 9 sites, whereas well-layered flows and one sill were sampled in the upper 17 sites. The flows are nearly flat lying.

At least four polarity reversals were recorded by the sequence. Within each polarity interval, several adjacent sites have statistically indistinguishable directions that were combined to derive unit means. Fifteen unit means and seven eruptive period means were derived from the sequence. The locality mean derived from the unit means is $D = 358°$, $I = 58°$ (Figure 15*e*, Table 4). The locality mean derived from the eruptive period means is $D = 4°$, $I = 57°$ (Figure 15*e*, Table 4).

Sou Hills. Samples were collected from basaltic flows and pyroclastic deposits and a rhyolite dome at 15 sites in the Sou Hills (SH, Figure 1). The rocks dip gently southeast.

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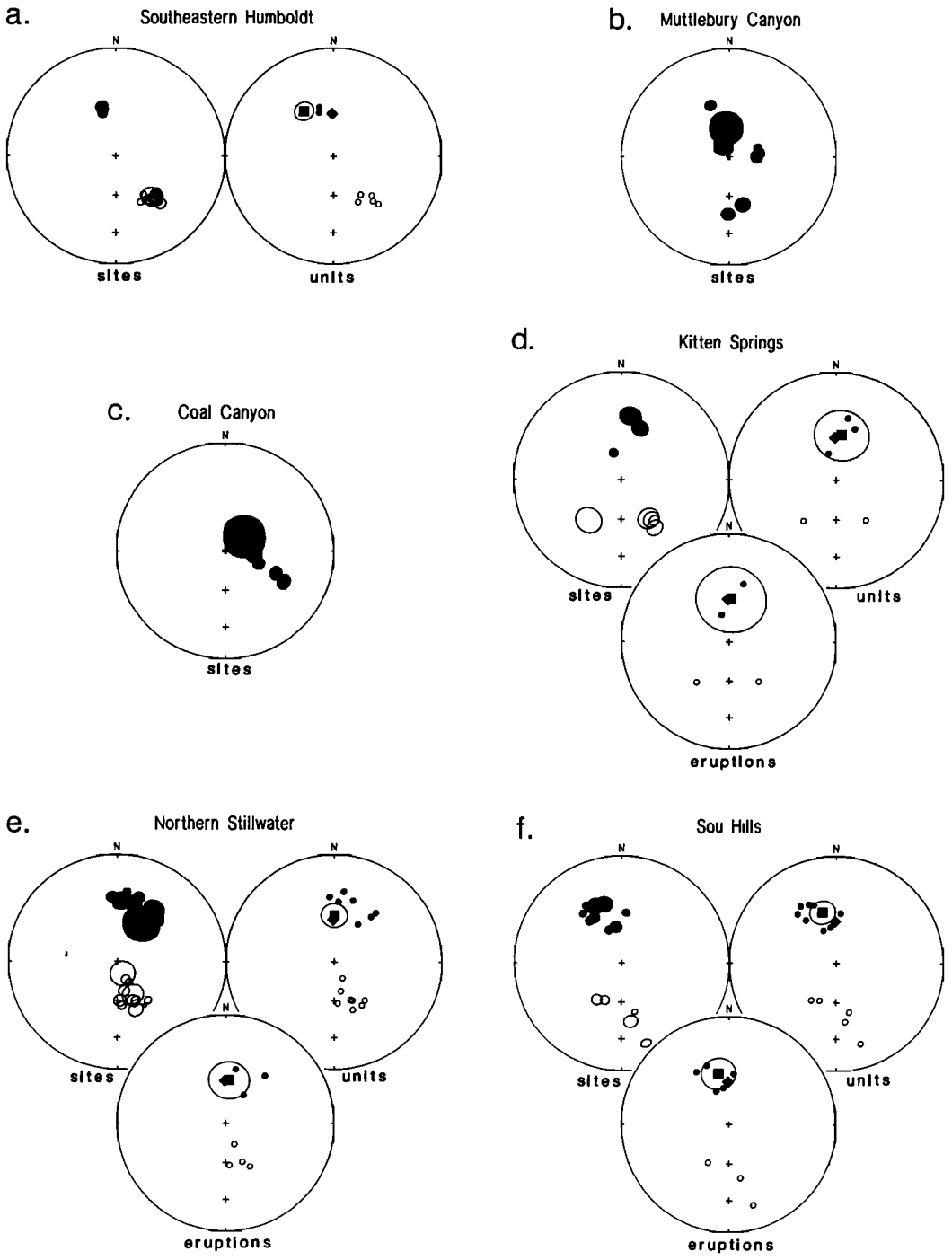


Fig. 15. Equal-area projections summarizing paleomagnetic data obtained at localities from middle Miocene basaltic rocks. Solid and open symbols are projections on lower and upper hemispheres, respectively. Cones of 95% confidence are shown for tilt-corrected site means for each locality. Circles depict tilt-corrected unit means or eruptive period means if four or more were derived for a locality. Squares, with cones of 95% confidence, represent locality means derived as an average of unit means or eruptive period means. Diamonds represent the time-average expected direction for the area.

Nosker [1981] obtained K/Ar ages of 17.5 to 15.0 Ma for the basalts and 14.4 Ma for the rhyolite dome. Relative ages between rocks sampled at some sites are uncertain because several normal faults offset the sequence.

Depending on the stratigraphic position assumed between rocks sampled in different fault blocks, either three or five changes of polarity were recorded by the sequence. A locality mean calculated from 13 unit means is $D = 345°$, $I = 49°$ (Figure 15*f*, Table 4). A locality mean derived from eight eruptive period means is $D = 348°$, $I = 50°$ (Figure 15*f*, Table 4).

Composite Means of Basalts

Paleomagnetic data from localities in the middle Miocene basalts were combined to more closely constrain an estimate

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TABLE 4. Middle Miocene Basalts, Locality Means and Rotation Estimates

Data Type	N	D	I	a_{95}	k	ASD	R	F
			Southeastern Humboldt Range					
Unit means	7	327	49	7.1	73	13	−31 ± 9	9 ± 6
			Kitten Spring Road					
Unit means	5	7	55	19.9	16	32	8 ± 28	3 ± 15
Eruptive period means	4	3	57	25.9	14	35	5 ± 41	0 ± 20
			Northern Stillwater Range					
Unit means	15	0	54	9.5	17	28	1 ± 13	4 ± 8
Eruptive period means	7	4	57	14.6	18	30	5 ± 22	1 ± 12
			Sou Hills					
Unit means	13	345	49	9.3	21	23	−13 ± 12	9 ± 7
Eruptive period means	8	348	50	12.4	21	21	−10 ± 15	8 ± 10
			Composite					
Unit means	43	349	53	5.2	18	27	−9 ± 7	4 ± 4
Eruptive period means*	23	353	55	7.1	19	27	−6 ± 10	8 ± 6
Eruptive period means†	20	354	56	7.6	19	26	−4 ± 11	2 ± 6

N, number of data used to calculate locality mean; D and I, declination and inclination of locality mean; a_{95}, 95% confidence limits about mean; k, precision parameter [*Fisher*, 1953]; ASD, angular standard deviation of virtual geomagnetic poles; R and F, rotation and flattening parameters (*Beck* [1980], with error limits after *Demarest*, [1983]).

*,†Different combinations of composite eruptive period means, as discussed in text.

of the middle Miocene dipole field direction for the Dixie Valley region. Unlike individual ash-flow tuffs that typically have widespread distribution, it is unlikely that the same individual basalt flow or pyroclastic horizon was sampled at more than one locality. However, sequences of basalts may have been emplaced during identical discrete eruptive periods at more than one locality. With this possibility in mind, data from the basalts were combined in three ways to calculate composite time-averaged means. First, independent unit means from all localities were combined, excluding all sites from the Coal Canyon locality and the lower six sites from the Muttlebury Canyon locality that are believed to record transitional fields. For 43 independent cooling units (Figure 16*a*), after tilt correction, the composite mean is D = 349°, I = 53°. In the second calculation, eruptive period means from each locality (Figure 16*b*) were given equal weight. For 23 eruptive period means the composite mean is D = 353°, I = 55°. Finally, eruptive period means derived from potentially coeval basalt sequences among the localities were first combined before all eruptive period means were combined. Eruptive sequences that were suspected to be coeval were a reverse polarity (D = 137°–150°, I = −46° to − 52°) sequence of thin well-layered basalt flows present at the stratigraphically highest positions of the southeastern Humboldt, Kitten Springs, and northern Stillwater localities and a normal polarity (D = 10°–14°, I = 40°–44°) sequence of flows present at intermediate stratigraphic positions of the northern Stillwater and Kitten Spring localities. Data from these sequences were grouped into two eruptive means and combined with 18 other eruptive means (Figure 16*c*) to derive a time-averaged mean of D = 354°, I = 56°.

Rotation Estimates

Rotation parameters calculated from means for the four locality means and the composite means from the middle Miocene rocks are both positive and negative in sign yet typically of small magnitude (≤13°, Table 4). None of the rotation values calculated from eruptive period means is statistically significant (Figure 17, Table 4). We conclude that the basalts have not experienced significant counterclockwise vertical-axis rotation in comparison to Oligocene to lower Miocene rocks in the central part of the study area. For example, R values from the northern Stillwater locality (1° ± 13° from unit means; 5° ± 22° from eruptive period

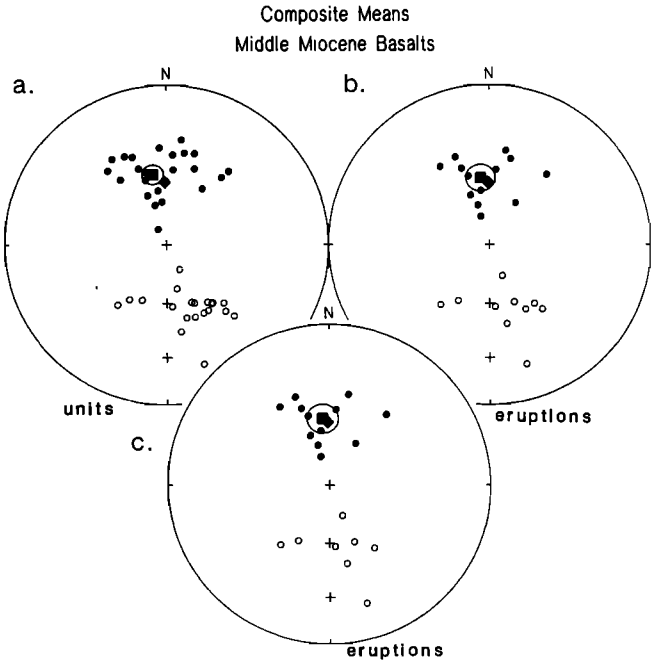


Fig. 16. Equal-area projections depicting tilt-corrected paleomagnetic data (circles) used to derive composite means (squares and associated cones of 95% confidence) for middle Miocene basaltic rocks from the Dixie Valley region. Solid and open symbols are projections on lower and upper hemispheres, respectively. (*a*) Forty-three unit means. (*b*) Twenty-three eruptive period means. (*c*) Twenty eruptive period means derived by combining data from potentially coeval eruptive periods at different localities (see text for more detail). Diamonds represent the time-averaged expected direction for the area.

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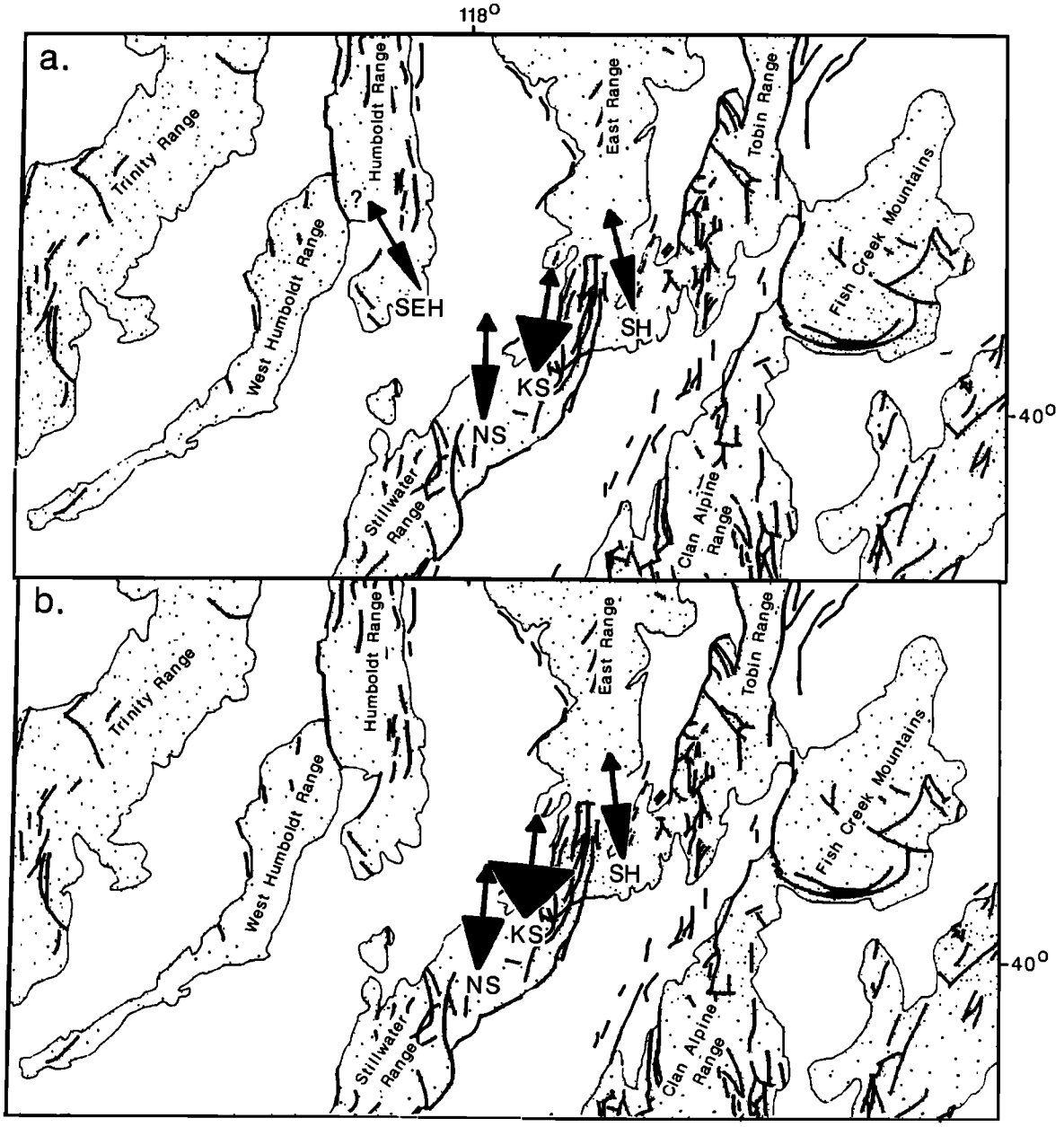


Fig. 17. Maps of the northern half of the study area showing estimates of vertical-axis rotation derived from locality means of paleomagnetic data from middle Miocene basaltic rocks. Rotation magnitude and sense are depicted relative to present geographic north by arrows. Angles subtended by arrows are 95% confidence limits for rotation estimates (calculated using the method of *Demarest* [1983]). KS, Kitten Springs; NS, northern Stillwater; SEH, southeastern Humboldt; SH, Sou Hills. (*a*) Rotation estimates derived from averages of unit means. Queried estimate for southeastern Humboldt locality is probably less reliable due to inadequate averaging of geomagnetic secular variation by paleomagnetic data from this locality. (*b*) Rotation estimates derived from averages of eruptive period means.

means) are statistically insignificant and much smaller than the R value ($-28° \pm 20°$) derived from Oligocene to lower Miocene rocks at the White Rock Canyon locality only 10 km to the south. Whether the middle Miocene rocks experienced any counterclockwise vertical-axis rotation is less certain.

Statistically significant R values based on unit means for the southeast Humboldt Range and Sou Hills localities and for the composite mean may in part reflect inadequate averaging of secular variation rather than true counterclockwise vertical-axis rotation. Only the R value ($-31° \pm 9°$) for the southeast Humboldt Range locality is of large magnitude. Several observations suggest that secular variation has not been adequately averaged by the time-average mean for this locality: (1) low angular standard deviation of VGPs (3° and 6°) indicates that flows within two polarity intervals were probably deposited in only two brief eruptive periods, (2) total angular standard deviation of VGPs for the seven unit means (13°) from the locality is about half that observed from other localities and the composite sequence (Table 4), (3) directions of normal- and reverse-polarity means are not statistically antipodal, and (4) the flattening value for the locality is statistically significant (9° + 6°). The southeast Humboldt Range locality probably has not been affected by

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a strong counterclockwise vertical-axis rotation since middle Miocene time.

Rotation values that slightly exceed statistical confidence limits derived from unit means from the Sou Hills (−13° ± 12°) and unit means of the composite sequence (−9° ± 7°) could be interpreted to suggest that the area experienced a small amount of counterclockwise rotation since middle Miocene time. In contrast to data from the southeast Humboldt Range, angular standard deviation of VGPs indicate that unit means from the Sou Hills (23°) and the composite sequence (27°) recorded considerable dispersion of the geomagnetic field. However, flattening values that exceed or equal their confidence limits for the Sou Hills (9° ± 7°) and the composite sequence (4° ± 4°) suggest that averages of the unit means still are not completely accurate estimates of time-averaged dipole field directions and their associated confidence limits. Rotation and flattening values derived from the eruptive period means in these two cases are not statistically significant, and rotation values derived from the eruptive period means are 3°–5° less than estimates from the unit means (Table 4). As discussed earlier, time-averaged locality means or composite means may be biased by data from groups of unit means that record limited secular variation within one or more eruptive periods. Unit means from eruptive periods whose directions are counterclockwise and shallow compared to a true dipole direction, such as at the southeast Humboldt Range locality, may have biased the Sou Hills and composite time-averaged mean and may have led to an underestimate of the statistical confidence limits for the rotation and flattening parameters. On the other hand, confidence limits for rotation and flattening parameters may be overestimated for means calculated from the eruptive period means because these average some secular variation.

Paleomagnetic data from the middle Miocene basalts indicate that significant Cenozoic vertical-axis rotation in the Dixie Valley region ceased before middle Miocene time. This is compatible with the contrast in the style and degree of deformation that affects the middle Miocene basaltic rocks and underlying Oligocene to lower Miocene ash-flow tuff sequence in areas such as the Stillwater Range.

Discussion and Conclusions

Paleomagnetic data from Tertiary igneous rocks of the Dixie Valley region document counterclockwise vertical-axis rotation in the area and bear on interpretation of the timing and spatial extent of the rotation. Oligocene to lower Miocene rocks from the central part of the study area experienced counterclockwise vertical-axis rotation (Figure 12) often exceeding 25°. Data from a composite stratigraphic section of the Oligocene to lower Miocene rocks from the central part of the area have declinations that, on average, shift counterclockwise with increasing age of units. This stratigraphically controlled shift of declinations implies that vertical-axis rotation was occurring during the time span (about 23–33 Ma) of ash-flow deposition. Paleomagnetic data indicate that middle Miocene basaltic rocks (about 10–17 Ma) were erupted after significant counterclockwise vertical-axis rotation (Figure 17), although interpretation of these data is complicated by an apparently episodic but rapid extrusion of the basalt flows. Thus the counterclockwise rotation is attributed to the older of two Tertiary deformation phases that have been recognized in the area.

The total amount and time of inception of counterclockwise rotation are not well constrained by paleomagnetic data from the Oligocene to lower Miocene rocks. Rotation estimates calculated from time-averaged means on the basis of data from a broad age range of Oligocene to lower Miocene units probably underestimate the total amount of Tertiary rotation, because they include data from units probably younger than much of the rotation. For example, the rotation estimate for the entire composite sequence from the central area is −23° ± 15°, whereas estimates from older and younger halves of the sequence are −37° ± 21° and −11° ± 16°, respectively. An estimate of rotation (−52° ± 12°) derived from paleomagnetic data from Jurassic volcanic rocks in the Stillwater Range [*Hudson*, 1988] may provide a better indication of maximum Tertiary rotation; its large magnitude suggests that rotation may have begun before deposition of the oldest Tertiary units.

Paleomagnetic data provide information on the spatial variability of counterclockwise vertical-axis rotation in the Dixie Valley region, but the age of the sampled units must be considered in such interpretations. The Golconda Canyon locality contains the oldest Tertiary units (≥33 Ma) at any locality sampled, therefore it should have experienced a larger rotation (greater than the 37° estimated for the older half of the composite sequence) if its deformation history was similar to localities from the central part of the area. Although poorly constrained, the low rotation estimate (−6° ± 32°) for the Golconda Canyon locality suggests that significant vertical-axis rotation did not occur in this area. Rocks sampled at the New Pass locality are the youngest of the Oligocene to lower Miocene sequence. The low rotation estimate for this locality (3° ± 37°) is not unreasonable, but it does not preclude middle Tertiary vertical-axis rotation prior to deposition of these units. For localities from the central part of the area that contain correlatable ash-flow tuffs, comparison of means from individual units suggests that only the western Stillwater locality experienced differential rotation with respect to other localities.

Paleomagnetic data from this study are insufficient to fully define the geographic extent of the area that experienced Tertiary counterclockwise vertical-axis rotation. However, these data and their comparison with other published paleomagnetic data do provide insight into the scale and timing of vertical-axis rotations within the Great Basin area of the Basin and Range province. For convenience, we consider three scales at which vertical-axis rotation may occur: (1) rotations of local extent that affect areas of a few square kilometers or less which may reflect drag along individual strike-slip or oblique-slip faults and local distortions of fault blocks, (2) rotations of regional extent that affect areas of tens to several hundred square kilometers and may involve a single rigid block or coherent systems of numerous fault blocks within a single domain [e.g., *Ron et al.*, 1984], or (3) rotations that are province-wide, affecting the entire Great Basin. It is improbable that the vertical-axis rotation detected in the Dixie Valley area is solely a local-scale phenomena; localities with statistically significant rotation estimates encompass a large area (about 40 × 50 km) and exhibit a consistent sense of rotation. The approximately 20° greater rotation inferred for the western Stillwater locality, however, could be due to locally developed structures. In contrast, the lack of rotation inferred for some localities (Golconda Canyon locality and perhaps the Eastgate–

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southern Desatoya locality) as well as at other locations in the Great Basin (e.g., the composite of data by *Gromme et al.* [1972], *Russell et al.* [1982], and *Geissman et al.* [1984]) indicate that Oligocene to early Miocene counterclockwise rotation did not affect the entire Great Basin province. These observations indicate that counterclockwise vertical-axis rotation in the Dixie Valley region occurs at a regional scale and affects a discrete but presently poorly defined area. Vertical-axis rotations of known or inferred Tertiary age have been detected in other areas in the Great Basin [*Gillett and Van Alstine*, 1982; *Nelson and Jones*, 1987; *Hagstrum and Gans*, 1989; *Geissman et al.*, 1989; *Li et al.*, 1990; *Rosenbaum et al.*, 1991], but these studies found that both clockwise and counterclockwise vertical-axis rotation and the affected rocks are as young as middle Miocene. Several studies have shown that rotations are spatially variable at regional and local scales [*Gillett and Van Alstine*, 1982; *Nelson and Jones*, 1987; *Hagstrum and Gans*, 1989; *Geissman et al.*, 1989; *Rosenbaum et al.*, 1991]. The different ages and spatial variability of vertical-axis rotations detected thus far lead us to conclude that Tertiary vertical-axis rotation in the Great Basin acted at regional as well as local scales but not at the province scale. This contrasts with the pervasive and consistent clockwise vertical-axis rotation of province scale that affected the Pacific Northwest [e.g., *Beck*, 1980; *Wells and Coe*, 1985; *Gromme et al.*, 1986; *Wells*, 1989] and southern California [e.g., *Luyendyk et al.*, 1980, 1985] that appear, at least in part, to be the result of plate margin interactions. The scale, spatial variability and mixed senses of domains of vertical-axis rotation in the Great Basin may characterize how vertical-axis rotation, if it occurs, affects extensional regimes. Similar spatial characteristics have been suggested from paleomagnetic studies in the Colorado River region of the Basin and Range province [*Wells and Hillhouse*, 1989], in the Rio Grande Rift [*Brown and Golombek*, 1985], and in Aegean areas [*Kissel and Laj*, 1988].

Although we believe that counterclockwise vertical-axis rotation documented in the Dixie Valley area was restricted to a discrete structural domain, the domain did not act as a single rigidly rotating block. At the White Rock Canyon locality, for example, faulting and tilting were concurrent with ash-flow deposition and vertical-axis rotation [*Hudson and Geissman*, 1987; *Hudson*, 1988], demonstrating that the domain was not internally rigid. Faults that now strike northwest (Figure 4) had right-lateral slip. As noted by *Hudson and Geissman* [1987], a model in which the counterclockwise rotation within the domain was accommodated by rotation of blocks and intervening right-lateral faults in a domino manner best fits both paleomagnetic and structural data from the area. This is similar to the block rotation models of *Ron et al.* [1984] or *McKenzie and Jackson* [1986]. A more detailed account of this model will be discussed in a separate paper.

Several areas in the Great Basin were deformed in Oligocene to early Miocene time. Most prominent and best documented are metamorphic core complexes of the northeastern Great Basin (Ruby–East Humboldt Range [*Snoke and Lush*, 1984; *Dallmeyer et al.*, 1986; *Dokka et al.*, 1986], Snake-Schell Creek-Egan Ranges [*Miller et al.*, 1983; *Gans and Miller*, 1983; *Gans et al.*, 1986; *Lee et al.*, 1987], and Albion-Grouse Creek-Raft River Ranges [*Compton et al.*, 1977; *Miller*, 1980; *Todd*, 1980; *Saltzer and Hodges*, 1988]). We suspect that there may be a causal relation between these highly extended areas in the northeastern Great Basin and coeval counterclockwise vertical-axis rotation in the Dixie Valley region, but this relation is presently equivocal, given current knowledge of Oligocene to early Miocene deformation in the Great Basin. Data presented in this paper, however, illustrate that Oligocene to early Miocene deformation was spread over a greater area than a single medial belt in the northeastern Great Basin.

Acknowledgments. This research was conducted as part of a dissertation by M. R. H. at the Colorado School of Mines. Funding for this research came from a grant from the donors of the Petroleum Research Fund of the American Chemical Society to J. W. G. and a Geological Society of America research grant to M. R. H. Discussions with R. L. Reynolds, J. G. Rosenbaum, C. S. Gromme, and D. A. John have been helpful throughout the research. D. A. John graciously provided much unpublished geologic data concerning the southern Stillwater Range. C. S. Gromme, G. J. Calderone, and B. P. Luyendyk provided helpful reviews of the manuscript. Any use of trade names is for descriptive purposes only and does not imply endorsement by the U.S. Geological Survey.

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J. W. Geissman, Department of Geology, University of New Mexico, Albuquerque, NM 87131.

M. R. Hudson, U.S. Geological Survey, Box 25046, MS 913, Denver, CO 80225.

(Received June 12, 1990;
revised October 12, 1990;
accepted November 2, 1990).

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